MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COMPARATOR, MAGNITUDE, DIGITAL, KEY CODE JDB

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic magnitude comparator module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.11 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be JDB.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/142-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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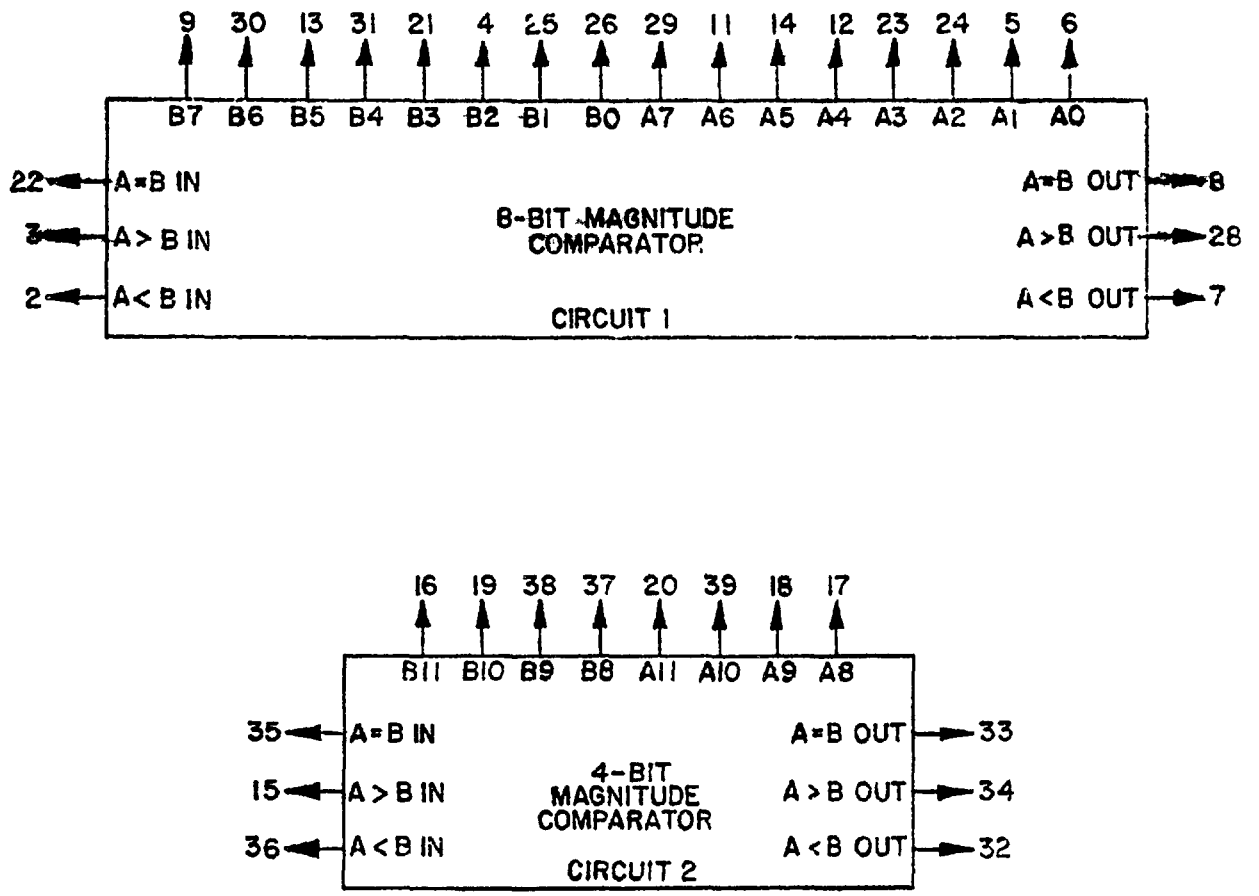


FIGURE 1. JDB module function diagram, magnitude comparator.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A6	V_{OUT} (low loaded)	III	S	10	800	10	800	10	800	10	800	mV
B1,B2	Isolation	IV	S	10	∞	1	∞	10	∞	1	∞	MΩ
C1-C30	Input breakdown current	V	S	0	1	0	1	0	1	0	1	mA
D1-D30	Input clamp diode voltage	VI	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
E1-E26	Input current, high level	VII	X	0	120	0	120	0	120	0	120	µA
E27-E30	Input current, high level	VII	X	0	40	0	40	0	40	0	40	µA
F1-F26	Input current, low level	VII	X	0.10	5.19	0.10	5.19	0.10	5.19	0.10	5.19	mA
F27-F30	Input current, low level	VII	X	0.10	1.73	0.10	1.73	0.10	1.73	0.10	1.73	mA
G1-G6	Output short circuit current	VII	X	20	55	20	55	20	55	20	55	mA
H1-H54	Functional compare	VIII	X	← See table VIII →								
J1-J6	Output voltage, high level	VIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
K1-K6	Output voltage, low level	VIII	X	0	400	0	400	0	400	0	400	mV
L	Power supply current	IX	X	50	265	50	265	50	265	50	265	mA
M1-M6	Transition time (t_{TLH}), C_L = 130 pF	X	S	5	45	5	45	5	50	5	50	ns
N1-N6	Transition time (t_{THL}), C_L = 130 pF	X	S	2	20	2	20	2	25	2	25	ns
P1-P4	Propagation delay time (t_{PLH}), C_L = 130 pF	XI	S	5	40	5	40	5	45	5	45	ns
P5	Propagation delay time (t_{PLH}), C_L = 130 pF	XI	S	5	50	5	50	5	55	5	55	ns

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See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P6-P9	Propagation delay time (t_{PLH}), C_L = 130 pF	XI	S	5	20	5	20	5	25	5	25	ns
P10	Propagation delay time (t_{PLH}), C_L = 130 pF	XI	S	5	30	5	30	5	35	5	35	ns
P11-P14, P19-P22, P27-P30, P35-P38	Propagation delay time (t_{PLH}), C_L = 130 pF	XI	S	5	35	5	35	5	40	5	40	ns
P15-P18, P23-P26, P31-P34, P39-P42	Propagation delay time (t_{PLH}), C_L = 130 pF	XI	S	5	45	5	45	5	50	5	50	ns
P43-P50, P59-P66, P75-P82, P91-P98	Propagation delay time (t_{PLH}), C_L = 130 pF	XI	S	5	55	5	55	5	60	5	60	ns
P51-P58, P67-P74, P83-P90, P99-P106	Propagation delay time (t_{PLH}), C_L = 130 pF	XI	S	5	65	5	65	5	70	5	70	ns
Q1-Q4	Propagation delay time (t_{PHL}), C_L = 130 pF	XI	S	5	45	5	45	5	50	5	50	ns
Q5	Propagation delay time (t_{PHL}), C_L = 130 pF	XI	S	5	50	5	50	5	55	5	55	ns
Q6-Q9	Propagation delay time (t_{PHL}), C_L = 130 pF	XI	S	5	25	5	25	5	30	5	30	ns
Q10	Propagation delay time (t_{PHL}), C_L = 130 pF	XI	S	5	30	5	30	5	35	5	35	ns

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See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q11-Q14, Q19-Q22, Q27-Q30, Q35-Q38	Propagation delay time (t_{PHL}), C_L = 130 pF	XI	S	5	40	5	40	5	45	5	45	ns
Q15-Q18, Q23-Q26, Q31-Q34, Q39-Q42	Propagation delay time (t_{PHL}), C_L = 130 pF	XI	S	5	40	5	40	5	45	5	45	ns
Q43-Q50, Q59-Q66, Q75-Q82, Q91-Q98	Propagation delay time (t_{PHL}), C_L = 130 pF	XI	S	5	60	5	60	5	65	5	65	ns
Q51-Q58, Q67-Q74, Q83-Q90, Q99-Q106	Propagation delay time (t_{PHL}), C_L = 130 pF	XI	S	5	60	5	60	5	65	5	65	ns
R	Power supply filter capacitance	XII	S	0.005	0.015	0.005	0.015	0.005	0.015	0.005	0.015	µF
S1-S6	Output crosstalk, high level	XIII	Q	0	400	0	400	0	400	0	400	mV
T1-T6	Output crosstalk, low level	XIII	Q	0	400	0	400	0	400	0	400	mV
U	Power supply transient amplitude	XIV	Q	0	400	0	400	0	400	0	400	mV
V1-V30	Input capacitance	XV	Q	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through V in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through U shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition R, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through U shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military Specification MIL-M-28787/142 supersedes Department of the Navy, Naval Electronic Systems Command Drawing 28687-0102-862. Part number M28787/142-1 supersedes part numbers 0102-862 and M28787/142.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Supply voltage	5.0 V dc ±10 percent
Maximum power dissipation	1.46 W (outputs not loaded)
Input voltage	
Low level	0.8 V maximum
High level	2.0 V minimum

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Input current (A, B, and A=B inputs)	
Low level (V_{IL} = 0.0 V)	-5.19 mA maximum
(V_{IL} = 0.4 V)	-4.8 mA maximum
High level (V_{IH} = 2.4 V)	120 µA maximum
(V_{IH} = 5.5 V)	1 mA maximum
Input current (A>B and A<B inputs)	
Low level (V_{IL} = 0.0 V)	-1.73 mA maximum
(V_{IL} = 0.4 V)	-1.6 mA maximum
High level (V_{IH} = 2.4 V)	40 µA maximum
(V_{IH} = 5.5 V)	1 mA maximum
Output voltage	
Low level (I_{OL} = 16.0 mA)	0.4 V maximum
High level (I_{OH} = -400 µA)	2.4 V minimum
Output drive current maximum capability	
Low level (V_{OL} = 0.4 V)	16 mA
High level (V_{OH} = 2.4 V)	-400 µA
Output short circuit current	55 mA maximum

Propagation delay time (C_L = 130 pF)

Circuit 1	Maximum limit	
	t_{PLH}	t_{PHL}
A<B IN to A>B OUT	45 ns	50 ns
A=B IN to A>B OUT	45 ns	50 ns
A>B IN to A<B OUT	45 ns	50 ns
A=B IN to A<B OUT	45 ns	50 ns
A=B IN to A=B OUT	55 ns	55 ns
A or B to A>B	60 ns	65 ns
A or B to A=B	70 ns	65 ns
A or B to A<B	60 ns	65 ns
Circuit 2		
A<B IN to A>B OUT	25 ns	30 ns
A=B IN to A>B OUT	25 ns	30 ns
A>B IN to A<B OUT	25 ns	30 ns
A=B IN to A<B OUT	25 ns	30 ns
A=B IN to A=B OUT	35 ns	35 ns
A or B to A>B	40 ns	45 ns
A or B to A=B	50 ns	45 ns
A or B to A<B	40 ns	45 ns

* 6.5 Module function. This module contains a four-bit and an eight-bit magnitude comparator. The four-bit comparator has four A and four B compare inputs and the eight-bit comparator has eight A and eight B compare inputs. Both comparators have A>B, A=B, and A<B outputs. Both comparators perform magnitude comparison of straight binary and straight BCD (8421) codes.

6.5.1 Input to output relationship. Each magnitude comparator module has the input to output relationship of table II for positive logic.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

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TABLE II. Input to output relationship. 1/ 3/

Comparing inputs												Cascading inputs			Outputs		
A11,B11	A10,B10	A9,B9	A8,B8	A7,B7	A6,B6	A5,B5	A4,B4	A3,B3	A2,B2	A1,B1	A0,B0	A>B	A<B	A=B	A>B	A<B	A=B
A11>B11	X	X	X									X	X	X	H	L	L
A11<B11	X	X	X									X	X	X	L	H	L
A11=B11	A10>B10	X	X									X	X	X	H	L	L
A11=B11	A10<B10	X	X									X	X	X	L	H	L
A11=B11	A10=B10	A9>B9	X									X	X	X	H	L	L
A11=B11	A10=B10	A9<B9	X									X	X	X	L	H	L
A11=B11	A10=B10	A9=B9	A8>B8									X	X	X	H	L	L
A11=B11	A10=B10	A9=B9	A8<B8									X	X	X	L	H	L
A11=B11	A10=B10	A9=B9	A8=B8									H	L	L	H	L	L
A11=B11	A10=B10	A9=B9	A8=B8									L	H	L	L	H	L
A11=B11	A10=B10	A9=B9	A8=B8									L	L	H	L	L	H
				A7>B7	X	X	X	X	X	X	X	X	X	X	H	L	L
				A7<B7	X	X	X	X	X	X	X	X	X	X	L	H	L
				A7=B7	A6>B6	X	X	X	X	X	X	X	X	X	H	L	L
				A7=B7	A6<B6	X	X	X	X	X	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5>B5	X	X	X	X	X	X	X	X	H	L	L
				A7=B7	A6=B6	A5<B5	X	X	X	X	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4>B4	X	X	X	X	X	X	X	H	L	L
				A7=B7	A6=B6	A5=B5	A4<B4	X	X	X	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3>B3	X	X	X	X	X	X	H	L	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3<B3	X	X	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2>B2	X	X	X	X	X	H	L	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2<B2	X	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1>B1	X	X	X	X	H	L	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1<B1	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1=B1	A0>B0	X	X	X	H	L	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1=B1	A0<B0	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1=B1	A0=B0	H	L	L	H	L	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1=B1	A0=B0	L	H	L	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1=B1	A0=B0	L	L	H	L	L	H

1/ H = high level.

2/ L = low level.

3/ X = don't care.

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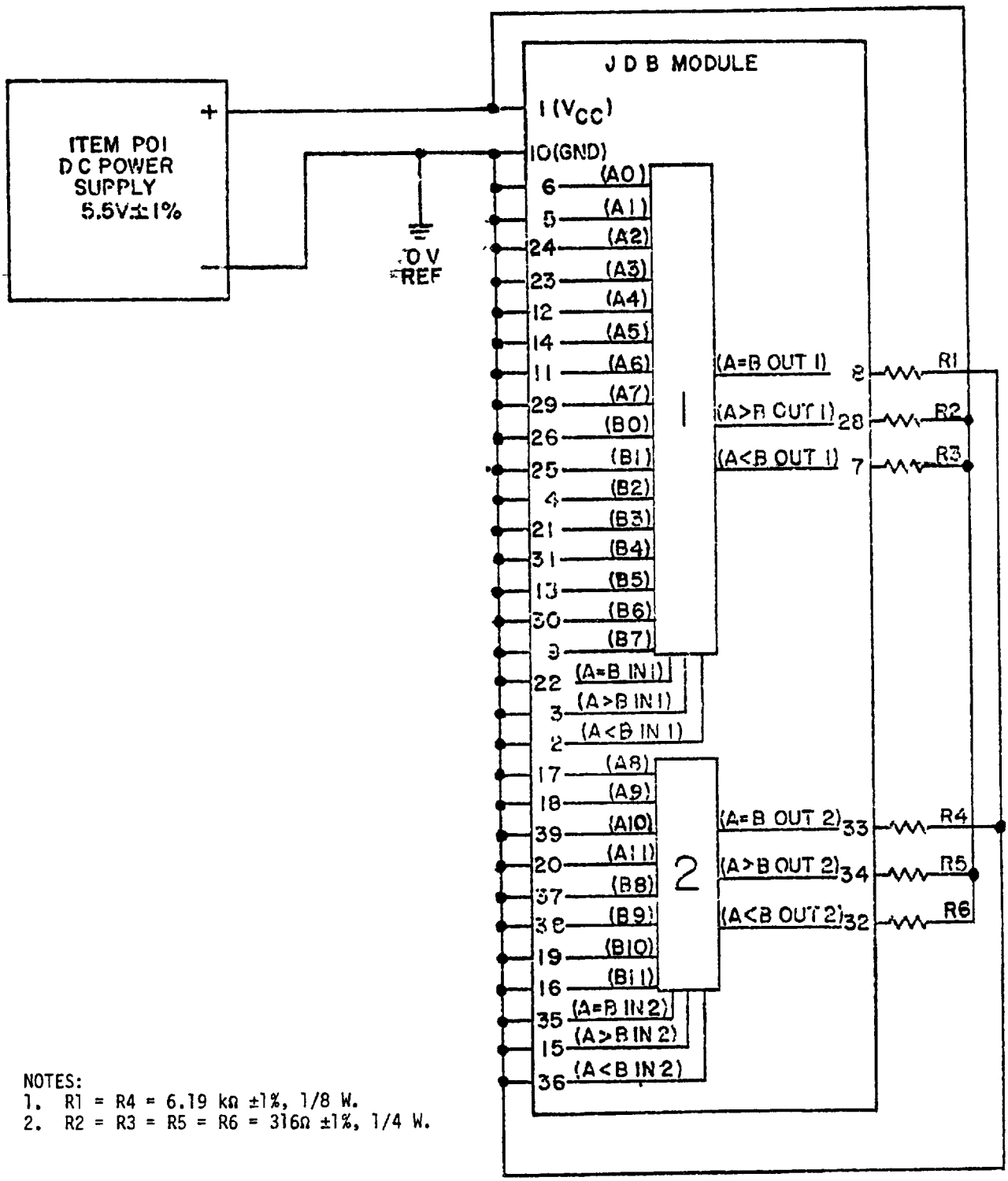
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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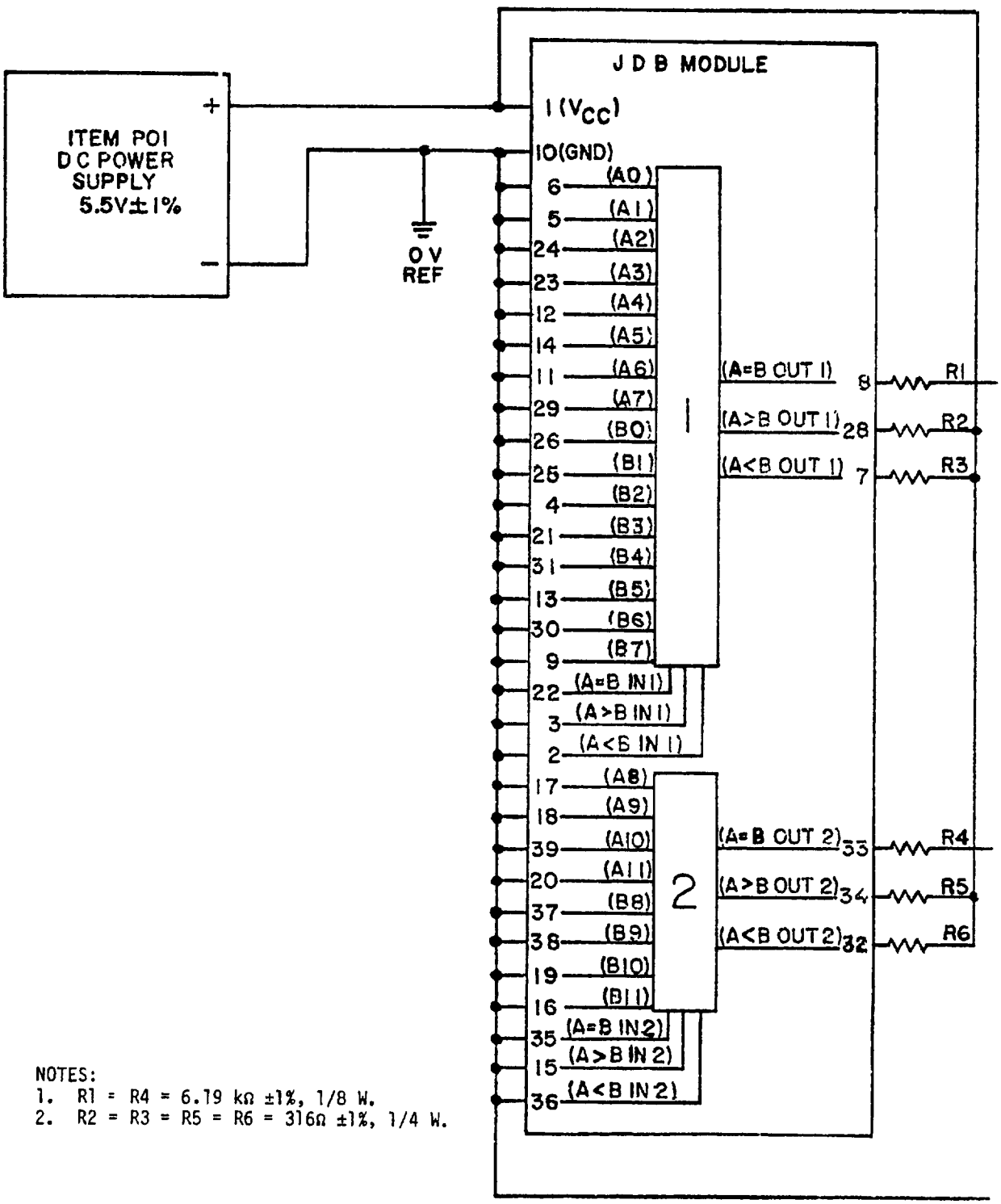
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NOTES:
1. R1 = R4 = 6.19 kΩ ±1%, 1/8 W.
2. R2 = R3 = R5 = R6 = 316Ω ±1%, 1/4 W.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = R4 = 6.19 kΩ ±1%, 1/8 W.
2. R2 = R3 = R5 = R6 = 316Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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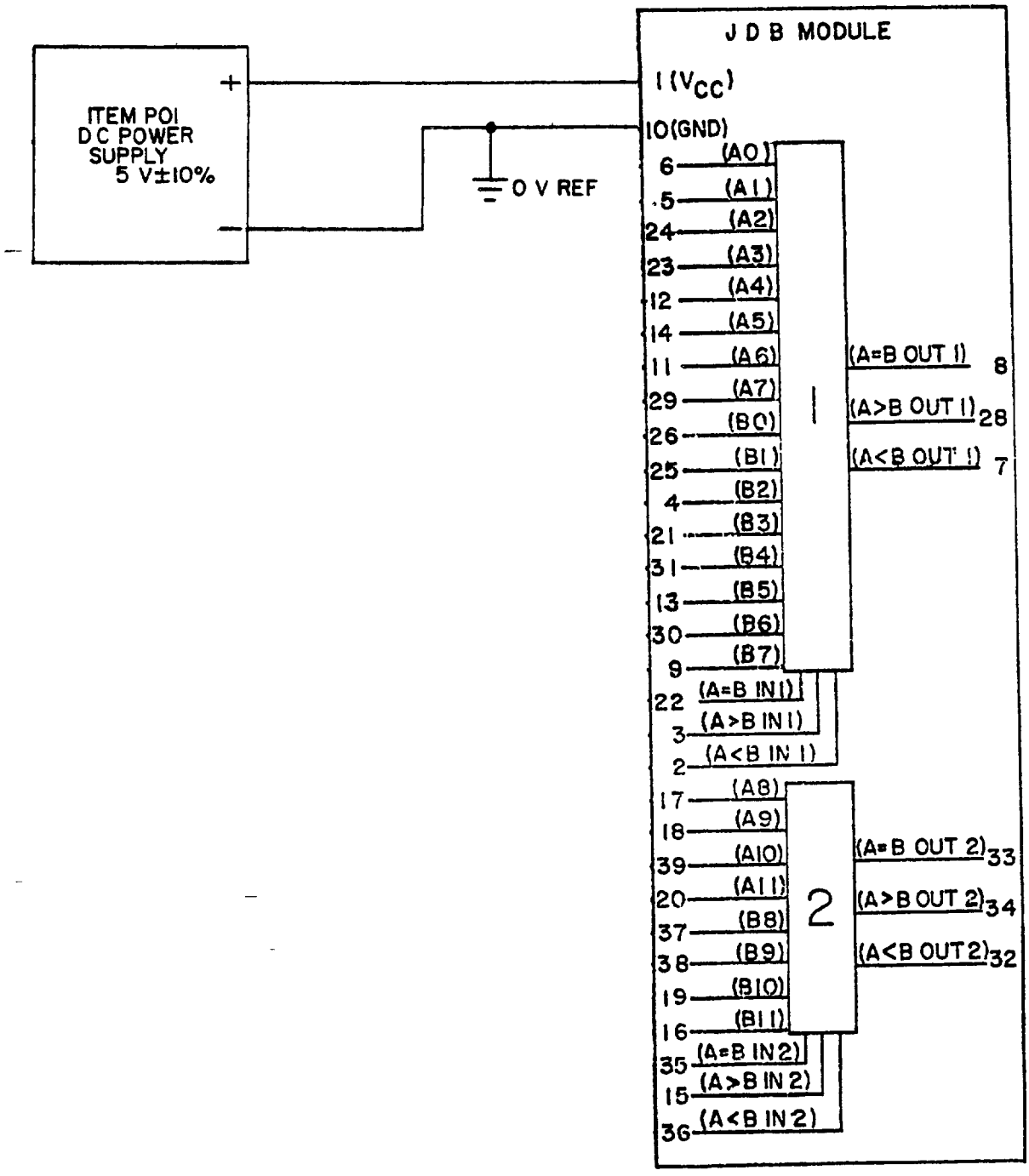


FIGURE 4. Vibration test circuit.

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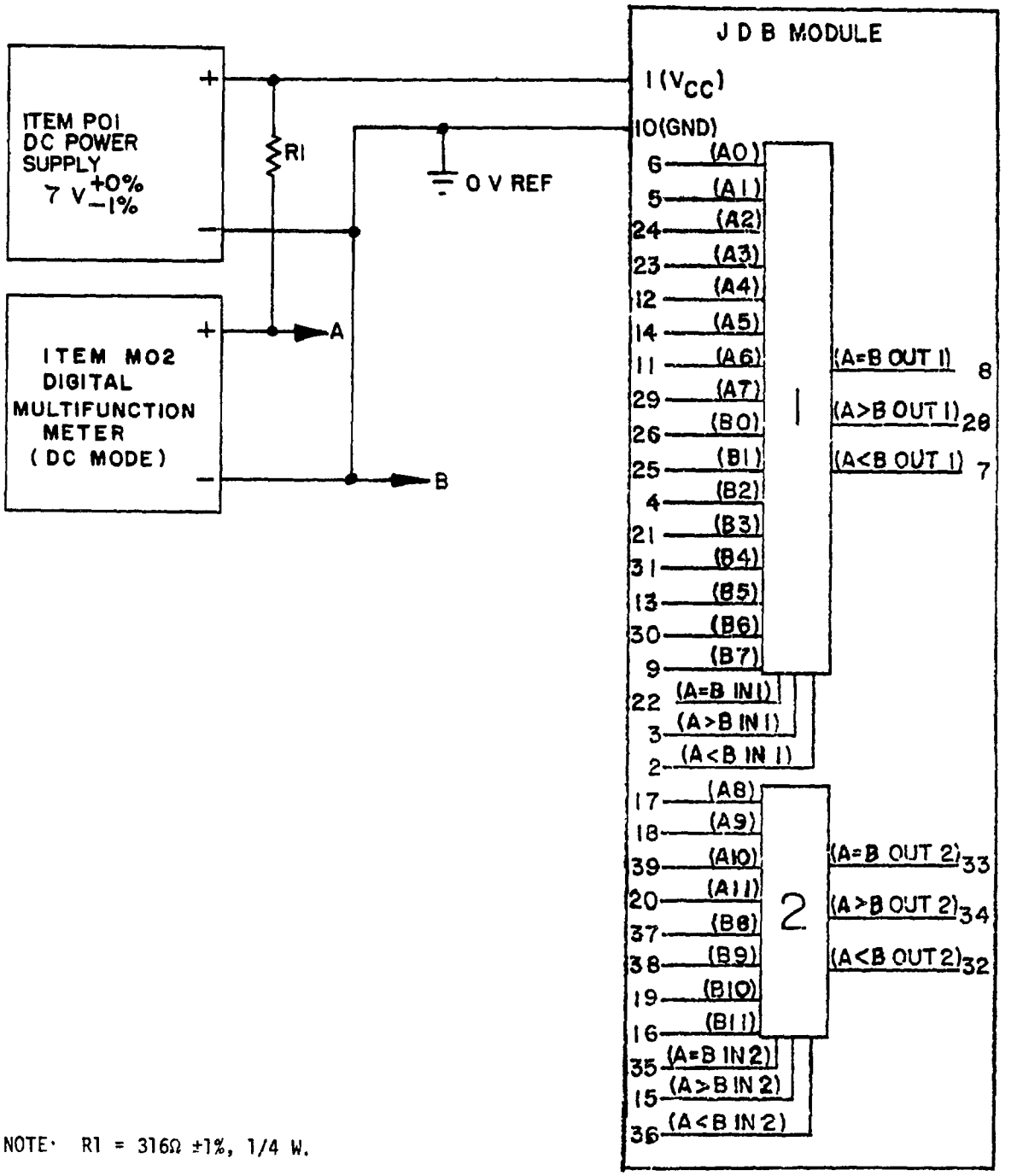
* TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	1A>B OUT V_{OUT} (low loaded)	28	29	10	800	mV	5
A2	1A<B OUT "	7	9	"	"	"	"
A3	1A=B OUT "	8	9	"	"	"	"
A4	2A>B OUT "	34	20	"	"	"	"
A5	2A<B OUT "	32	16	"	"	"	"
A6	2A=B OUT "	33	16	"	"	"	"

TABLE IV. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Isolation, frame to module pins	All pins	Frame	10	∞	MΩ	6
B2	Isolation, used to unused pins	All pins except 27,40	27,40	10	∞	MΩ	6

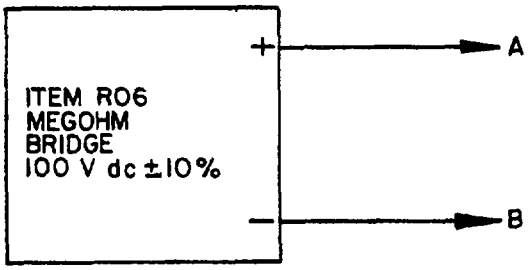
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:16Z
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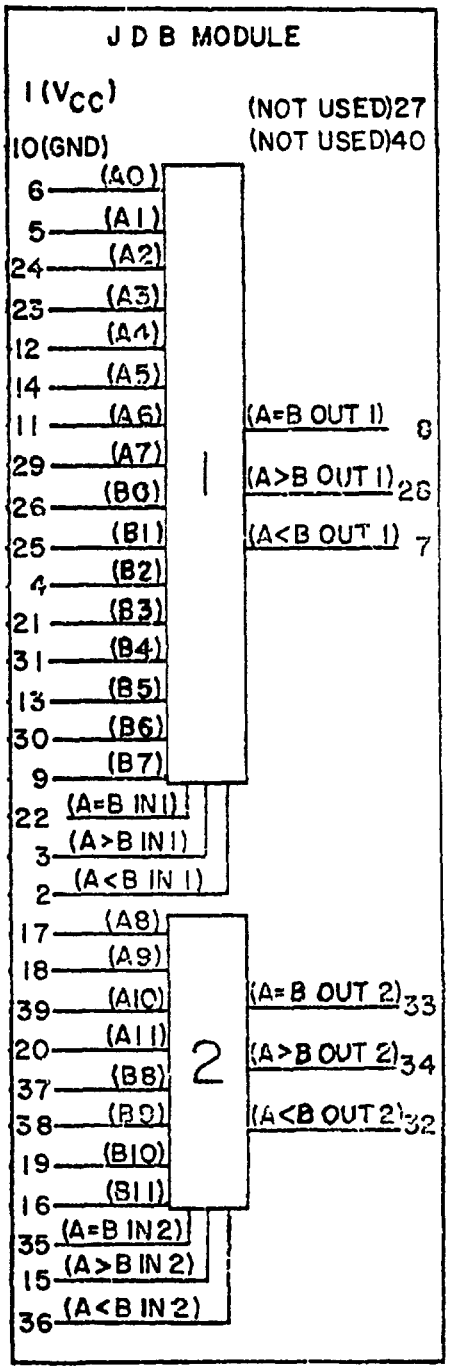


NOTE: R1 = 316Ω ±1%, 1/4 W.

* FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition B1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

* FIGURE 6. Isolation test circuit for table IV.

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TABLE V. Input breakdown current test.

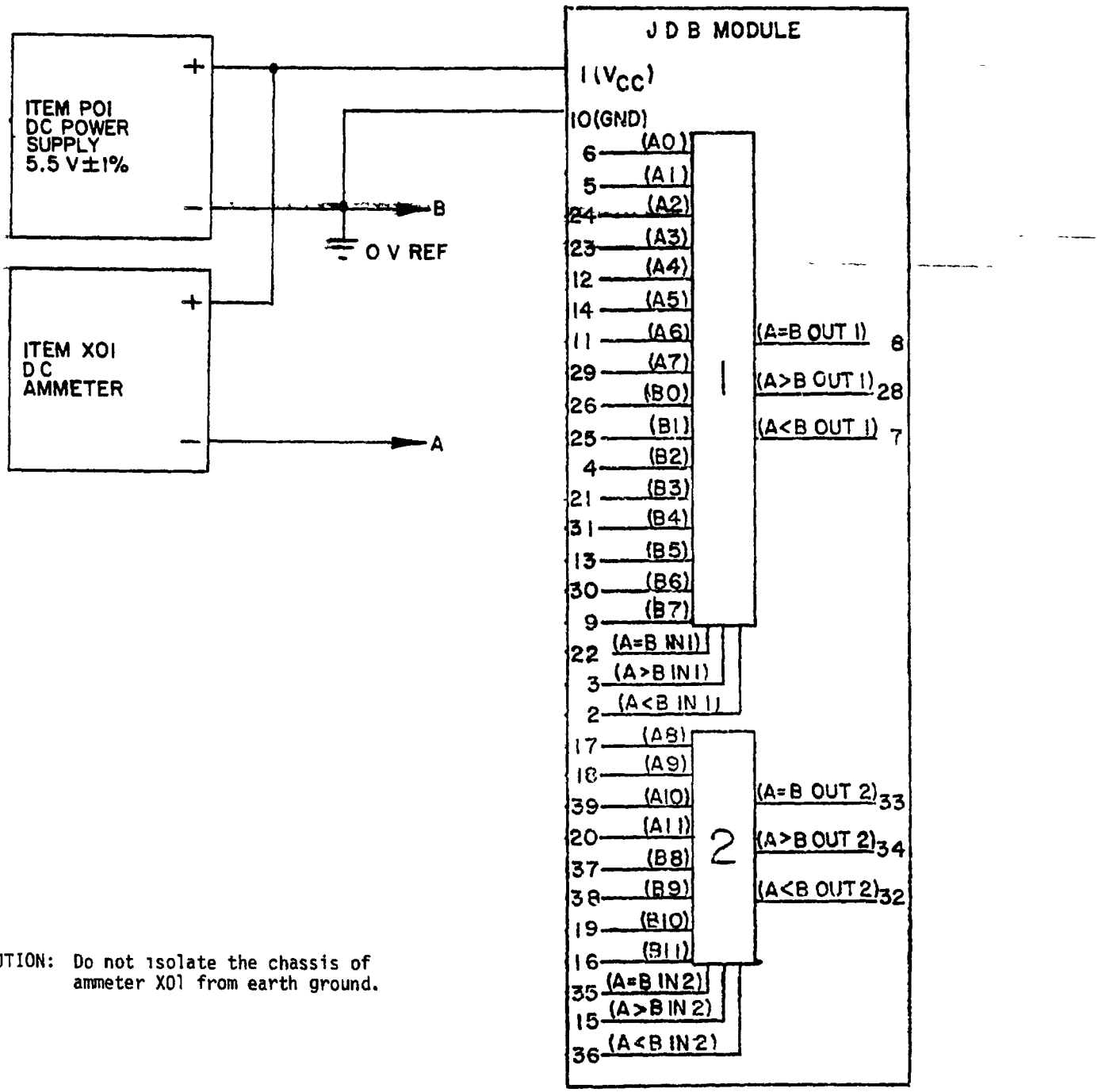
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	A0	Input breakdown current	6	26	0	1	mA	7
C2	A1	"	5	25	"	"	"	"
C3	A2	"	24	4	"	"	"	"
C4	A3	"	23	21	"	"	"	"
C5	A4	"	12	31	"	"	"	"
C6	A5	"	14	13	"	"	"	"
C7	A6	"	11	30	"	"	"	"
C8	A7	"	29	9	"	"	"	"
C9	A8	"	17	37	"	"	"	"
C10	A9	"	18	38	"	"	"	"
C11	A10	"	39	19	"	"	"	"
C12	A11	"	20	16	"	"	"	"
C13	B0	"	26	6	"	"	"	"
C14	B1	"	25	5	"	"	"	"
C15	B2	"	4	24	"	"	"	"
C16	B3	"	21	23	"	"	"	"
C17	B4	"	31	12	"	"	"	"
C18	B5	"	13	14	"	"	"	"
C19	B6	"	30	11	"	"	"	"
C20	B7	"	9	29	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C21	B8	Input breakdown current	37	17	0	1	mA	7
C22	B9	"	38	18	"	"	"	"
C23	B10	"	19	39	"	"	"	"
C24	B11	"	16	20	"	"	"	"
C25	1A=B IN	"	22	26	"	"	"	"
C26	1A>B IN	"	3	26	"	"	"	"
C27	1A<B IN	"	2	26	"	"	"	"
C28	2A=B IN	"	35	37	"	"	"	"
C29	2A>B IN	"	15	37	"	"	"	"
C30	2A<B IN	"	36	37	"	"	"	"

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CAUTION: Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 7. Input breakdown current test circuit for table V.

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* TABLE VI. Input clamp diode voltage test.

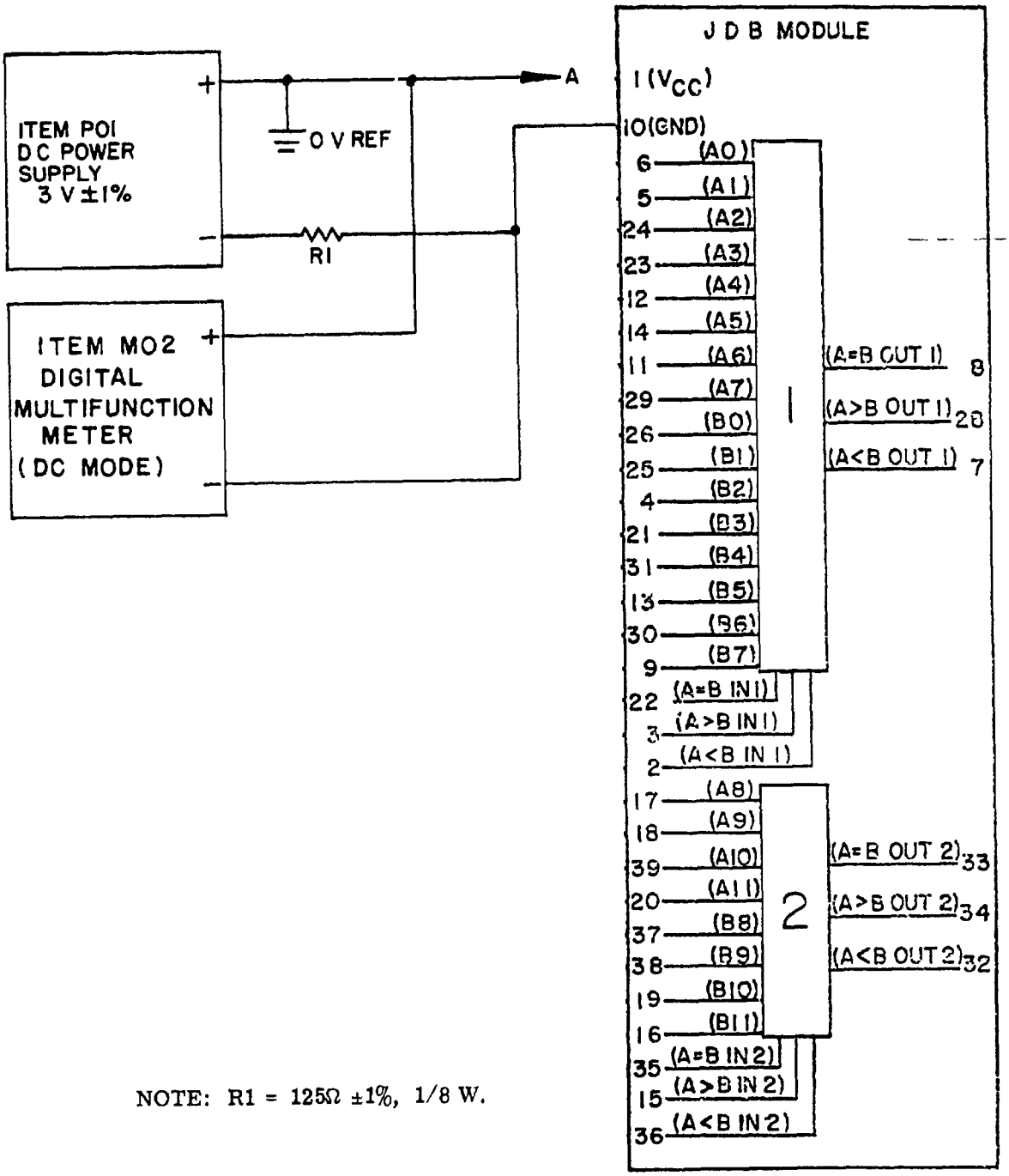
Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
D1	A0	Input clamp diode voltage	6	-0.4	-1.5	V	8
D2	A1	"	5	"	"	"	"
D3	A2	"	24	"	"	"	"
D4	A3	"	23	"	"	"	"
D5	A4	"	12	"	"	"	"
D6	A5	"	14	"	"	"	"
D7	A6	"	11	"	"	"	"
D8	A7	"	29	"	"	"	"
D9	A8	"	17	"	"	"	"
D10	A9	"	18	"	"	"	"
D11	A10	"	39	"	"	"	"
D12	A11	"	20	"	"	"	"
D13	B0	"	26	"	"	"	"
D14	B1	"	25	"	"	"	"
D15	B2	"	4	"	"	"	"
D16	B3	"	21	"	"	"	"
D17	B4	"	31	"	"	"	"
D18	B5	"	13	"	"	"	"
D19	B6	"	30	"	"	"	"
D20	B7	"	9	"	"	"	"

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* TABLE VI. Input clamp diode voltage test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
D21	B8	Input clamp diode voltage	37	-0.4	-1.5	V	8
D22	B9	"	38	"	"	"	"
D23	B10	"	19	"	"	"	"
D24	B11	"	16	"	"	"	"
D25	1A=B IN	"	22	"	"	"	"
D26	1A>B IN	"	3	"	"	"	"
D27	1A<B IN	"	2	"	"	"	"
D28	2A=B IN	"	35	"	"	"	"
D29	2A>B IN	"	15	"	"	"	"
D30	2A<B IN	"	36	"	"	"	"

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NOTE: R1 = 125Ω ±1%, 1/8 W.

*FIGURE 8. Input clamp diode voltage test circuit for table VI.

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TABLE VII. Input current (high-level and low-level) and output short circuit current test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
E1	A0	Input current, high level	6	Open	26	0	120	µA	9
E2	A1	"	5	"	25	"	"	"	"
E3	A2	"	24	"	4	"	"	"	"
E4	A3	"	23	"	21	"	"	"	"
E5	A4	"	12	"	31	"	"	"	"
E6	A5	"	14	"	13	"	"	"	"
E7	A6	"	11	"	30	"	"	"	"
E8	A7	"	29	"	9	"	"	"	"
E9	A8	"	17	"	37	"	"	"	"
E10	A9	"	18	"	38	"	"	"	"
E11	A10	"	39	"	19	"	"	"	"
E12	A11	"	20	"	16	"	"	"	"
E13	B0	"	26	"	6	"	"	"	"
E14	B1	"	25	"	5	"	"	"	"
E15	B2	"	4	"	24	"	"	"	"
E16	B3	"	21	"	23	"	"	"	"
E17	B4	"	31	"	12	"	"	"	"
E18	B5	"	13	"	14	"	"	"	"
E19	B6	"	30	"	11	"	"	"	"
E20	B7	"	9	"	29	"	"	"	"

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TABLE VII. Input current (high-level and low-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E21	B8 Input current, high level	37	Open	17	0	120	µA	9
E22	B9 "	38	"	18	"	"	"	"
E23	B10 "	19	"	39	"	"	"	"
E24	B11 "	16	"	20	"	"	"	"
E25	1A=B IN "	22	"	26	"	"	"	"
E26	2A=B IN "	35	"	37	"	"	"	"
E27	1A>B IN "	3	"	26	"	40	"	"
E28	1A<B IN "	2	"	26	"	"	"	"
E29	2A>B IN "	15	"	37	"	"	"	"
E30	2A<B IN "	36	"	37	"	"	"	"
F1	A0 Input current, low level	26	6	Open	0.10	5.19	mA	"
F2	A1 "	25	5	"	"	"	"	"
F3	A2 "	4	24	"	"	"	"	"
F4	A3 "	21	23	"	"	"	"	"
F5	A4 "	31	12	"	"	"	"	"
F6	A5 "	13	14	"	"	"	"	"
F7	A6 "	30	11	"	"	"	"	"
F8	A7 "	9	29	"	"	"	"	"
F9	A8 "	37	17	"	"	"	"	"
F10	A9 "	38	18	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:16Z
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TABLE VII. Input current (high-level and low-level) and output short circuit current test - Continued.

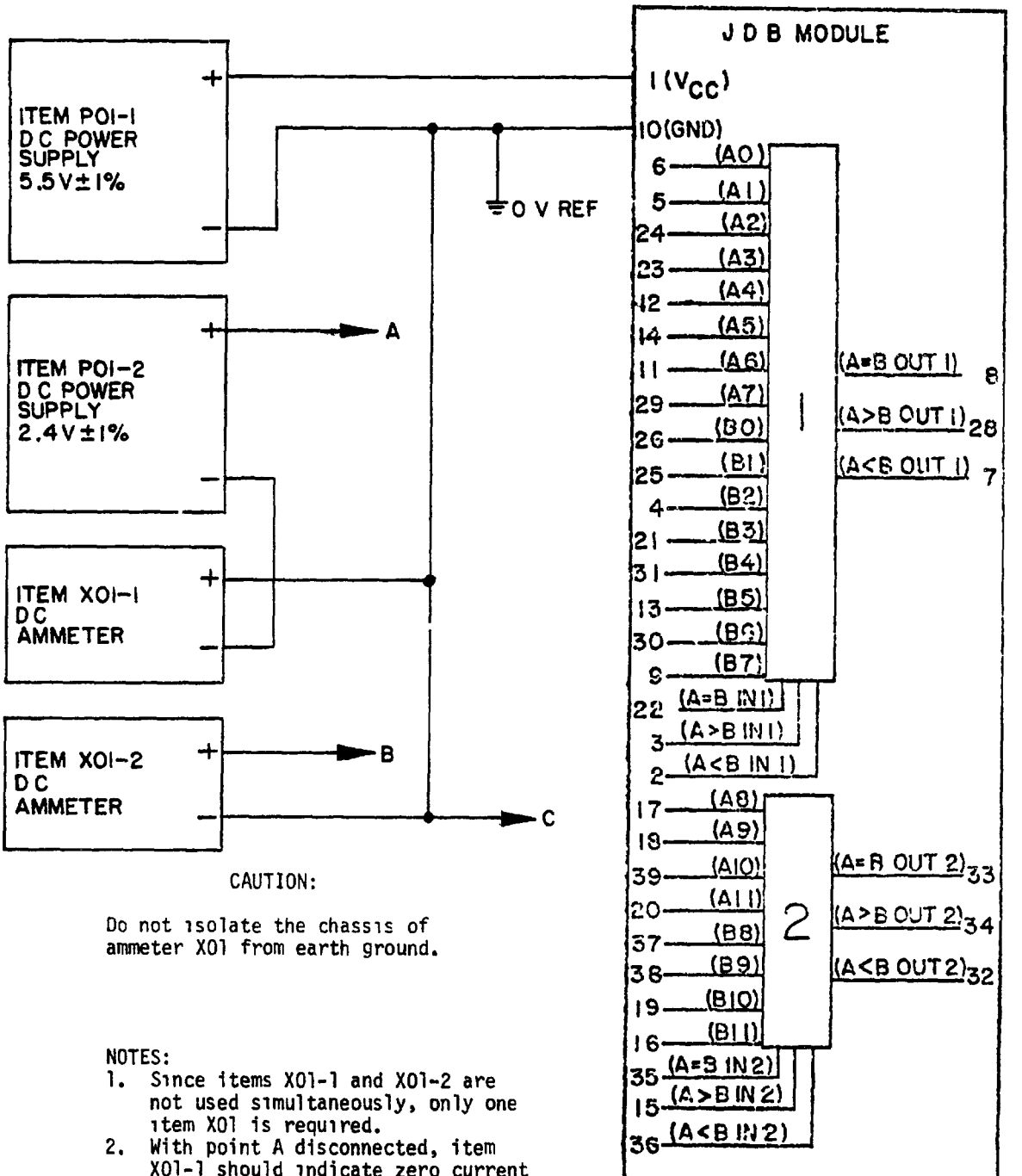
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F11	A10 Input current, low level	19	39	Open	0.10	5.19	mA	9
F12	A11 "	16	20	"	"	"	"	"
F13	B0 "	6	26	"	"	"	"	"
F14	B1 "	5	25	"	"	"	"	"
F15	B2 "	24	4	"	"	"	"	"
F16	B3 "	23	21	"	"	"	"	"
F17	B4 "	12	31	"	"	"	"	"
F18	B5 "	14	13	"	"	"	"	"
F19	B6 "	11	30	"	"	"	"	"
F20	B7 "	29	9	"	"	"	"	"
F21	B8 "	17	37	"	"	"	"	"
F22	B9 "	18	38	"	"	"	"	"
F23	B10 "	39	19	"	"	"	"	"
F24	B11 "	20	16	"	"	"	"	"
F25	1A=B IN "	Open	22	"	"	"	"	"
F26	2A=B IN "	"	35	"	"	"	"	"
F27	1A>B IN "	"	3	"	"	1.73	"	"
F28	1A<B IN "	"	2	"	"	"	"	"
F29	2A>B IN "	"	15	"	"	"	"	"
F30	2A<B IN "	"	36	"	"	"	"	"

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TABLE VII. Input current (high-level and low-level) and output short circuit current test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G1	1A=B OUT	Output short circuit current	22	8	2,3	20	55	mA	9
G2	1A>B OUT	"	3	28	2,22	"	"	"	"
G3	1A<B OUT	"	2	7	3,22	"	"	"	"
G4	2A=B OUT	"	35	33	15,36	"	"	"	"
G5	2A>B OUT	"	15	34	35,36	"	"	"	"
G6	2A<B OUT	"	36	32	15,35	"	"	"	"

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.
2. With point A disconnected, item X01-1 should indicate zero current on the 1.0 µA range.
3. When performing output short circuit current test, only one output shall be shorted at a time, and the measurement taken within 10 seconds.

FIGURE 9. Input current (high-level and low-level) and output short circuit current test circuit for table VII.

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* TABLE VIII. Functional compare and output voltage (high-level and low-level) test.

Test condition	Parameter	Connect					Outputs 1/			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	A=B	A>B	A<B	Min	Max		
H1	Functional compare test	15,35,36	20	16	2/	Open	L	H	L	1/		- -	10
H2	"	"	16	20	"	"	L	L	H	"		"	"
H3	"	"	16,20	Open	"	"	H	L	L	"		"	"
H4	"	"	Open	16,20	"	"	H	L	L	"		"	"
H5	"	"	39	19	"	16,20	L	H	L	"		"	"
H6	"	"	19	39	"	"	L	L	H	"		"	"
H7	"	"	19,39	Open	"	"	H	L	L	"		"	"
H8	"	"	Open	19,39	"	"	H	L	L	"		"	"
H9	"	"	18	38	"	16,19 20,39	L	H	L	"		"	"
H10	"	"	38	18	"	"	L	L	H	"		"	"
H11	"	"	18,38	Open	"	"	H	L	L	"		"	"
H12	"	"	Open	18,38	"	"	H	L	L	"		"	"
H13	"	"	17	37	"	16,18, 19,20, 38,39	L	H	L	"		"	"
H14	"	"	37	17	"	"	L	L	H	"		"	"
H15	"	"	17,37	Open	"	"	H	L	L	"		"	"
H16	"	"	Open	17,37	"	"	H	L	L	"		"	"
H17	"	Open	35	15,36	"	16,17, 18,19, 20,37, 38,39	H	L	L	"		"	"

See footnotes at end of table.

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* TABLE VIII. Functional compare and output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect					Outputs 1/			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	A=B	A>B	A<B	Min	Max		
H18	Functional compare test	Open	15	35,36	2/	16,17, 18,19, 20,37, 38,39	L	H	L	1/		- -	10
H19	"	Open	36	15,35	"	16,17, 18,19, 20,37, 38,39	L	L	H	"		"	"
H20	"	2,3,22	29	9	"	Open	L	H	L	"		"	"
H21	"	"	9	29	"	"	L	L	H	"		"	"
H22	"	"	9,29	Open	"	"	H	L	L	"		"	"
H23	"	"	Open	9,29	"	"	H	L	L	"		"	"
H24	"	"	11	30	"	9,29	L	H	L	"		"	"
H25	"	"	30	11	"	"	L	L	H	"		"	"
H26	"	"	11,30	Open	"	"	H	L	L	"		"	"
H27	"	"	Open	11,30	"	"	H	L	L	"		"	"
H28	"	"	14	13	"	9,11, 29,30	L	H	L	"		"	"
H29	"	"	13	14	"	"	L	L	H	"		"	"
H30	"	"	13,14	Open	"	"	H	L	L	"		"	"
H31	"	"	Open	13,14	"	"	H	L	L	"		"	"
H32	"	"	12	31	"	9,11, 13,14, 29,30	L	H	L	"		"	"

See footnotes at end of table.

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* TABLE VIII. Functional compare and output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect					Outputs 1/			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	A=B	A>B	A<B	Min	Max		
H33	Functional compare test	2,3,22	31	12	2/	9,11, 13,14, 29,30	L	L	H	1/		- -	10
H34	"	"	12,31	Open	"	"	H	L	L	"		"	"
H35	"	"	Open	12,31	"	"	H	L	L	"		"	"
H36	"	"	23	21	"	9,11, 12,13, 14,29, 30,31	L	H	L	"		"	"
H37	"	"	21	23	"	"	L	L	H	"		"	"
H38	"	"	21,23	Open	"	"	H	L	L	"		"	"
H39	"	"	Open	21,23	"	"	H	L	L	"		"	"
H40	"	"	24	4	"	9,11, 12,13, 14,21, 23,29, 30,31	L	H	L	"		"	"
H41	"	"	4	24	"	"	L	L	H	"		"	"
H42	"	"	4,24	Open	"	"	H	L	L	"		"	"
H43	"	"	Open	4,24	"	"	H	L	L	"		"	"

See footnotes at end of table.

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* TABLE VIII. Functional compare and output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect					Outputs 1/			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	A=B	A>B	A<B	Min	Max		
H44	Functional compare test	2,3,22	5	25	2/	4,9, 11,12, 13,14, 21,23, 24,29, 30,31	L	H	L	1/		- -	10
H45	"	"	25	5	"	"	L	L	H	"		"	"
H46	"	"	5,25	Open	"	"	H	L	L	"		"	"
H47	"	"	Open	5,25	"	"	H	L	L	"		"	"
H48	"	"	6	26	"	4,5,9, 11,12, 13,14, 21,23, 24,25, 29,30, 31	L	H	L	"		"	"
H49	"	"	26	6	"	"	L	L	H	"		"	"
H50	"	"	6,26	Open	"	"	H	L	L	"		"	"
H51	"	"	Open	6,26	"	"	H	L	L	"		"	"
H52	"	Open	22	2,3	"	4,5,6,9, 11,12, 13,14, 21,23, 24,25, 26,29, 30,31	H	L	L	"		"	"
H53	"	"	3	2,22	"	"	L	H	L	"		"	"
H54	"	"	2	3,22	"	"	L	L	H	"		"	"

See footnotes at end of table.

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* TABLE VIII. Functional compare and output voltage (high-level and low-level) test - Continued.

Test condition	Parameter		Connect					Outputs 1/			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	A=B	A>B	A<B	Min	Max		
J1	1A=B OUT	Output voltage, high level	Open	22	Open	8	2,3,16, 17,18, 19,20, 37,38,39	H	L	L	2.4	4.5	V	10
J2	1A>B OUT	"	"	3	"	28	2,22	L	H	L	"	"	"	"
J3	1A<B OUT	"	"	2	"	7	3,16,17, 18,19, 20,22, 37,38,39	L	L	H	"	"	"	"
J4	2A=B OUT	"	"	35	"	33	4,5,6,9, 11,12,13, 14,15,21, 23,24,25, 26,29,30, 31,36	H	L	L	"	"	"	"
J5	2A>B OUT	"	"	15	"	34	35,36	L	H	L	"	"	"	"
J6	2A<B OUT	"	"	36	"	32	4,5,6,9, 11,12, 13,14, 15,21, 23,24, 25,26, 29,30, 31,35	L	L	H	"	"	"	"
K1	1A=B OUT	Output voltage, low level	2	Open	22	8	3,16,17, 18,19, 20,37, 38,39	L	L	H	0	400	mV	"
K2	1A>B OUT	Output voltage, low level	22	Open	3	28	2	H	L	L	0	400	mV	"

See footnotes at end of table.

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* TABLE VIII. Functional compare and output voltage (high-level and low-level) test - Continued.

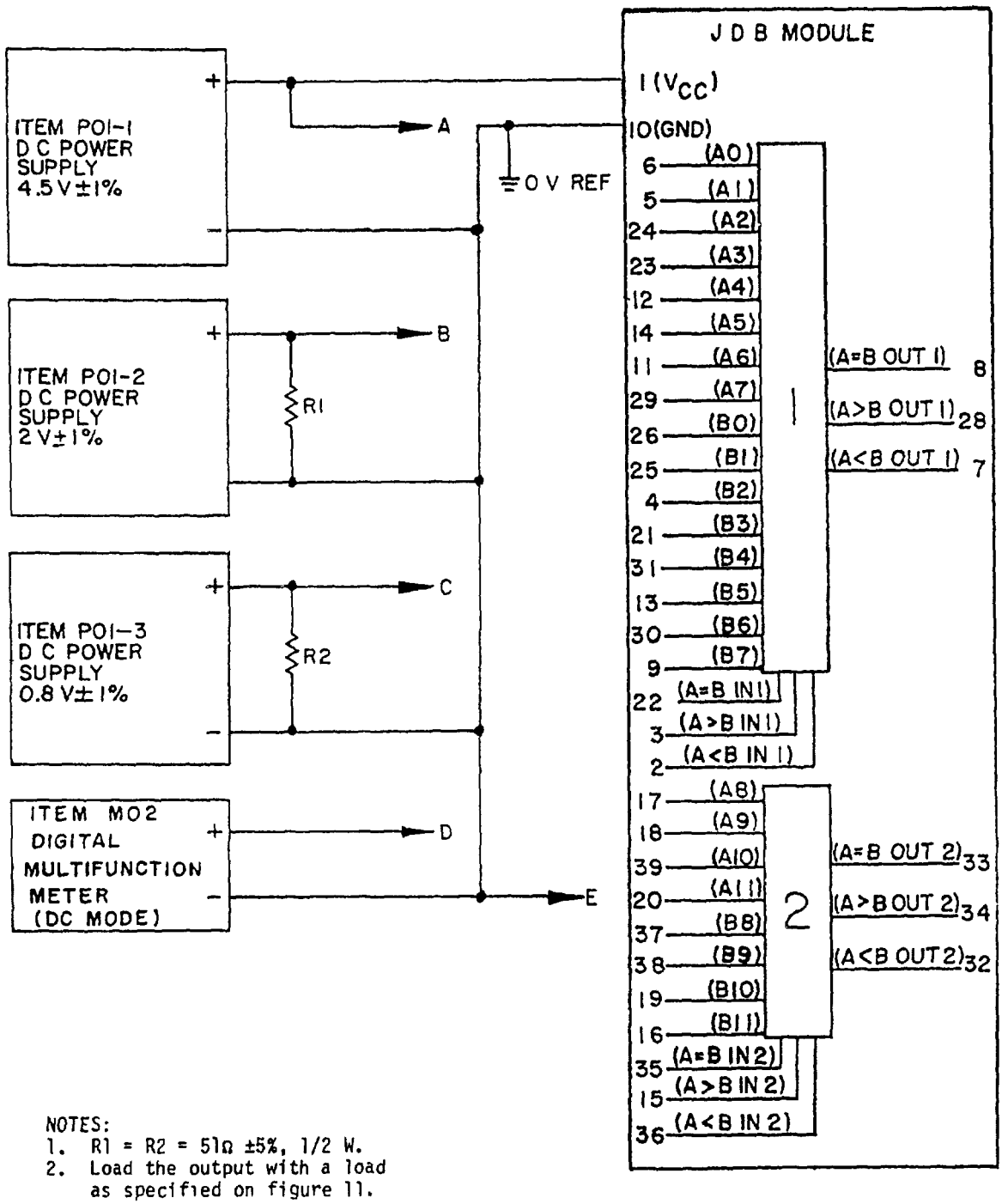
Test condition	Parameter		Connect					Outputs 1/			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	A=B	A>B	A<B	Min	Max		
K3	1A<B OUT	Output voltage, low level	3	Open	2	7	16,17, 18,19, 20,22, 37,38,39	L	H	L	0	400	mV	10
K4	2A=B OUT	"	36	"	35	33	4,5,6,9, 11,12, 13,14, 15,21, 23,24, 25,26, 29,30,31	L	L	H	"	"	"	"
K5	2A>B OUT	"	35	"	15	34	4,5,6,9, 11,12, 13,14, 21,23, 24,25, 26,29, 30,31,36	H	L	L	"	"	"	"
K6	2A<B OUT	"	15	"	36	32	4,5,6, 9,11, 12,13, 14,21, 23,24, 25,26, 29,30, 31,35	L	H	L	"	"	"	"

1/ Output logic states:

L = low level < 0.4 V.
H = high level > 2.4 V.

2/ For test condition H, verification of the output states specified in footnote 1 is obtained by connecting item M02 (point D) to each output pin after input conditions are accomplished.

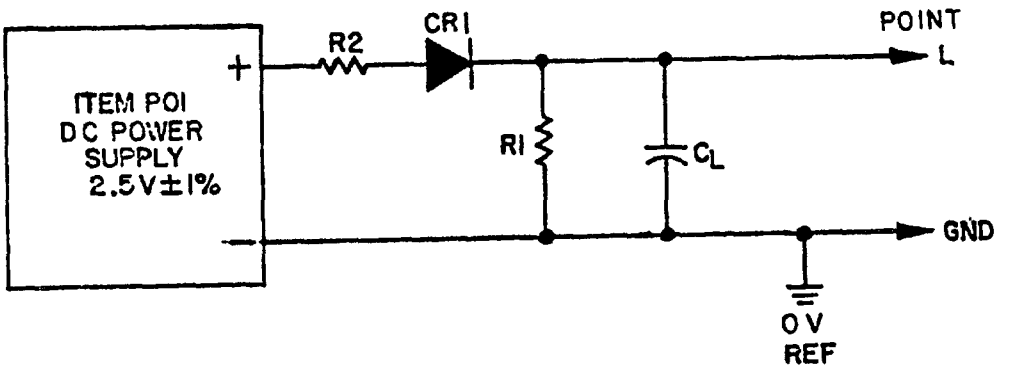
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load the output with a load as specified on figure 11.

* FIGURE 10. Functional compare and output voltage (high-level and low-level) test circuit for table VIII.

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NOTES:
1. R1 = 6.19 kΩ ±1%, 1/8 W.
2. R2 = Select value for a current flow of 16 mA ±1% with point L held at 0.4 V.
3. CR1 = 1N4150, or equivalent.
4. For test conditions M, N, P, and Q, C_L = 130 ±5 pF (including probe and parasitic capacitance).
5. For test conditions S and T, C_L = 50 ±5 pF (including probe and parasitic capacitance).
6. C_L is not required for dc electrical tests H, J, and K.

FIGURE 11. Logic load circuit.

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TABLE IX. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
L	Power supply current	50	265	mA	12

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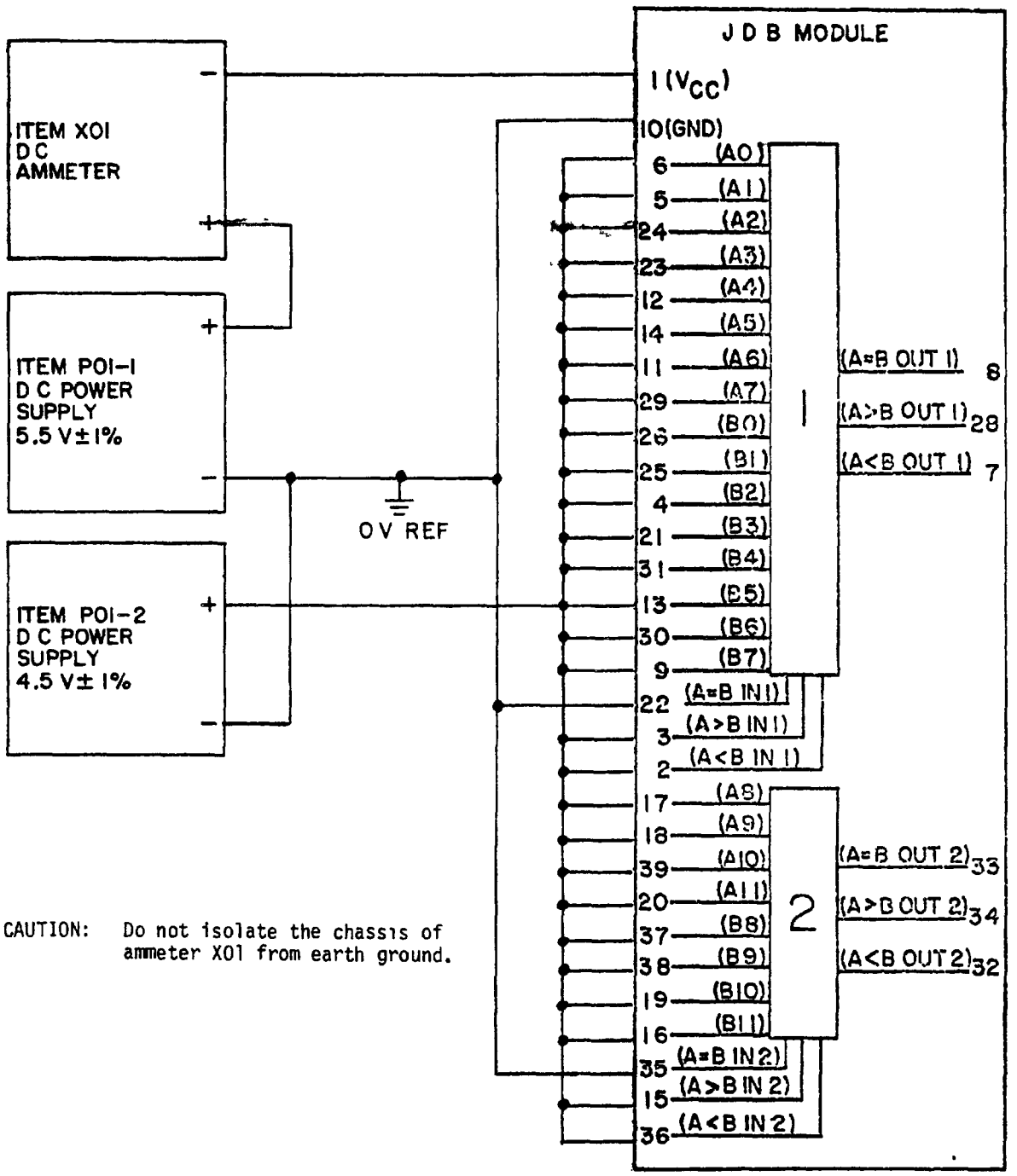


FIGURE 12. Power supply current test circuit for table IX.

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* TABLE X. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
M1	1A>B OUT	Transition time (t_{TLH}), C_L = 130 pF	Open	3	22	28	2,4,5,6,9, 11,12,13,14, 21,23,24,25, 26,29,30,31	5	45	ns	13,14
M2	1A<B OUT	"	"	2	22	7	3	"	"	"	"
M3	1A=B OUT	"	"	22	3	8	2,4,5,6,9, 11,12,13,14, 21,23,24,25, 26,29,30,31	"	"	"	"
M4	2A>B OUT	"	"	15	35	34	16,17,18,19, 20,36,37,38, 39	"	"	"	"
M5	2A<B OUT	"	"	36	35	32	15	"	"	"	"
M6	2A=B OUT	"	"	35	15	33	16,17,18,19, 20,36,37,38, 39	"	"	"	"
N1	1A>B OUT	Transition time (t_{THL}), C_L = 130 pF	"	3	22	28	2,4,5,6,9, 11,12,13,14, 21,23,24,25, 26,29,30,31	2	20	"	"
N2	1A<B OUT	"	"	2	22	7	3	"	"	"	"
N3	1A=B OUT	"	"	22	3	8	2,4,5,6,9, 11,12,13,14, 21,23,24,25, 26,29,30,31	"	"	"	"
N4	2A>B OUT	"	"	15	35	34	16,17,18,19, 20,36,37,38, 39	"	"	"	"
N5	2A<B OUT	"	"	36	35	32	15	"	"	"	"
N6	2A=B OUT	"	"	35	15	33	16,17,18,19, 20,36,37,38, 39	"	"	"	"

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* TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
P1	1A<B IN to 1A>B OUT Propagation delay time (t_{PLH}), C_L = 130 pF	Open	2	3	28	4,5,6,9,11, 12,13,14,21, 22,23,24,25, 26,29,30,31	5	40	ns	13,14
P2	1A=B IN to 1A>B OUT "	"	22	3	28	2	"	"	"	"
P3	1A>B IN to 1A<B OUT "	"	3	2	7	22	"	"	"	"
P4	1A=B IN to 1A<B OUT "	"	22	2	7	3	"	"	"	"
P5	1A=B IN to 1A=B OUT "	"	22	2	8	3,4,5,6,9, 11,12,13,14, 21,23,24,25, 26,29,30,31	"	50	"	"
P6	2A<B IN to 2A>B OUT "	"	36	15	34	16,17,18,19, 20,35,37,38, 39	"	20	"	"
P7	2A=B IN to 2A>B OUT "	"	35	15	34	36	"	"	"	"
P8	2A>B IN to 2A<B OUT "	"	15	36	32	35	"	"	"	"
P9	2A=B IN to 2A<B OUT "	"	35	36	32	15	"	"	"	"
P10	2A=B IN to 2A=B OUT "	"	35	36	33	15,16,17,18, 19,20,37,38, 39	"	30	"	"
P11	A11>B11 Propagation delay time (t_{PLH}), pulse on A, 0 V on B, C_L = 130 pF	"	20	Open	34	16	"	35	"	"
P12	A10>B10 Propagation delay time (t_{PLH}), pulse on A, 0 V on B, C_L = 130 pF	"	39	Open	34	16,19,20	"	35	"	"

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* TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
P13	A9>B9	Propagation delay time (t_{PLH}), pulse on A, 0 V on B, C_L = 130 pF	Open	18	Open	34	16,19,20,38, 39	5	35	ns	13,14
P14	A8>B8	"	"	17	"	34	16,18,19,20, 37,38,39	"	35	"	"
P15	A11=B11	"	"	20	"	33	16	"	45	"	"
P16	A10=B10	"	"	39	"	"	16,19,20	"	"	"	"
P17	A9=B9	"	"	18	"	"	16,19,20, 38,39	"	"	"	"
P18	A8=B8	"	"	17	"	"	16,18,19,20, 37,38,39	"	"	"	"
P19	A11<B11	Propagation delay time (t_{PLH}), pulse on A, 5 V on B, C_L = 130 pF	16	20	"	32	Open	"	35	"	"
P20	A10<B10	"	16,19,20	39	"	"	"	"	"	"	"
P21	A9<B9	"	16,19,20,38, 39	18	"	"	"	"	"	"	"
P22	A8<B8	"	16,18,19,20, 37,38,39	17	"	"	"	"	"	"	"
P23	A11=B11	"	16	20	"	33	"	"	45	"	"
P24	A10=B10	"	16,19,20	39	"	"	"	"	"	"	"
P25	A9=B9	"	16,19,20,38, 39	18	"	"	"	"	"	"	"
P26	A8=B8	"	16,18,19,20, 37,38,39	17	"	"	"	"	"	"	"
P27	A11>B11	Propagation delay time (t_{PLH}), pulse on B, 5 V on A, C_L = 130 pF	20	16	"	34	"	"	35	"	"
P28	A10>B10	Propagation delay time (t_{PLH}), pulse on B, 5 V on A, C_L = 130 pF	16,20,39	19	"	34	"	"	35	"	"

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* TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
P29	A9>B9	Propagation delay time (t_{PLH}), pulse on B, 5 V on A, C_L = 130 pF	16,18,19,20, 39	38	Open	34	Open	5	35	ns	13,14
P30	A8>B8	"	16,17,18,19, 20,38,39	37	"	34	"	"	35	"	"
P31	A11=B11	"	20	16	"	33	"	"	45	"	"
P32	A10=B10	"	16,20,39	19	"	"	"	"	"	"	"
P33	A9=B9	"	16,18,19,20, 39	38	"	"	"	"	"	"	"
P34	A8=B8	"	16,17,18,19, 20,38,39	37	"	"	"	"	"	"	"
P35	A11<B11	Propagation delay time (t_{PLH}), pulse on B, 0 V on A, C_L = 130 pF	Open	16	"	32	20	"	35	"	"
P36	A10<B10	"	"	19	"	"	16,20,39	"	"	"	"
P37	A9<B9	"	"	38	"	"	16,18,19,20, 39	"	"	"	"
P38	A8<B8	"	"	37	"	"	16,17,18,19, 20,38,39	"	"	"	"
P39	A11=B11	"	"	16	"	33	20	"	45	"	"
P40	A10=B10	"	"	19	"	"	16,20,39	"	"	"	"
P41	A9=B9	"	"	38	"	"	16,18,19,20, 39	"	"	"	"
P42	A8=B8	"	"	37	"	"	16,17,18,19, 20,38,39	"	"	"	"
P43	A7>B7	Propagation delay time (t_{PLH}), pulse on A, 0 V on B, C_L = 130 pF	"	29	"	28	9	"	55	"	"

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* TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
P44	A6>B6	Propagation delay time (t_{PLH}), pulse on A, 0 V on B, C_L = 130 pF	Open	11	Open	28	9,29,30	5	55	ns	13,14
P45	A5>B5	"	"	14	"	"	9,11,13,29, 30	"	"	"	"
P46	A4>B4	"	"	12	"	"	9,11,13,14, 29,30,31	"	"	"	"
P47	A3>B3	"	"	23	"	"	9,11,12,13, 14,21,29, 30,31	"	"	"	"
P48	A2>B2	"	"	24	"	"	4,9,11,12, 13,14,21,23, 29,30,31	"	"	"	"
P49	A1>B1	"	"	5	"	"	4,9,11,12, 13,14,21,23, 24,25,29,30, 31	"	"	"	"
P50	A0>B0	"	"	6	"	"	4,5,9,11,12, 13,14,21,23, 24,25,26,29, 30,31	"	"	"	"
P51	A7=B7	"	"	29	"	8	9	"	65	"	"
P52	A6=B6	"	"	11	"	"	9,29,30	"	"	"	"
P53	A5=B5	"	"	14	"	"	9,11,13,29, 30	"	"	"	"
P54	A4=B4	"	"	12	"	"	9,11,13,14, 29,30,31	"	"	"	"
P55	A3=B3	"	"	23	"	"	9,11,12,13, 14,21,29,30, 31	"	"	"	"

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* TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			.. to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
P56	A2=B2	Propagation delay time (t_{PLH}), pulse on A, 0 V on B, C_L = 130 pF	Open	24	Open	8	4,9,11,12, 13,14,21,23, 29,30,31	5	65	ns	13,14
P57	A1=B1	"	"	5	"	"	4,9,11,12, 13,14,21,23, 24,25,29,30, 31	"	"	"	"
P58	A0=B0	"	"	6	"	"	4,5,9,11,12, 13,14,21,23, 24,25,26,29, 30,31	"	"	"	"
P59	A7<B7	Propagation delay time (t_{PLH}), pulse on A, 5 V on B, C_L = 130 pF	9	29	"	7	Open	"	55	"	"
P60	A6<B6	"	9,29,30	11	"	"	"	"	"	"	"
P61	A5<B5	"	9,11,13,29, 30	14	"	"	"	"	"	"	"
P62	A4<B4	"	9,11,13,14, 29,30,31	12	"	"	"	"	"	"	"
P63	A3<B3	"	9,11,12,13, 14,21,29,30, 31	23	"	"	"	"	"	"	"
P64	A2<B2	"	4,9,11,12, 13,14,21,23, 29,30,31	24	"	"	"	"	"	"	"
P65	A1<B1	"	4,9,11,12, 13,14,21,23, 24,25,29,30, 31	5	"	"	"	"	"	"	"

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* TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
P66	A0<B0	Propagation delay time (t_{PLH}), pulse on A, 5 V on B, C_L = 130 pF	4,5,9,11,12, 13,14,21,23, 24,25,26,29, 30,31	6	Open	7	Open	5	55	ns	13,14
P67	A7=B7	"	9	29	"	8	"	"	65	"	"
P68	A6=B6	"	9,29,30	11	"	"	"	"	"	"	"
P69	A5=B5	"	9,11,13,29, 30	14	"	"	"	"	"	"	"
P70	A4=B4	"	9,11,13,14, 29,30,31	12	"	"	"	"	"	"	"
P71	A3=B3	"	9,11,12,13, 14,21,29, 30,31	23	"	"	"	"	"	"	"
P72	A2=B2	"	4,9,11,12, 13,14,21,23, 29,30,31	24	"	"	"	"	"	"	"
P73	A1=B1	"	4,9,11,12, 13,14,21, 23,24,25, 29,30,31	5	"	"	"	"	"	"	"
P74	A0=B0	"	4,5,9,11, 12,13,14, 21,23,24, 25,26,29, 30,31	6	"	"	"	"	"	"	"
P75	A7>B7	Propagation delay time (t_{PLH}), pulse on B, 5 V on A, C_L = 130 pF	29	9	"	28	"	"	55	"	"
P76	A6>B6	"	9,11,29	30	"	"	"	"	"	"	"
P77	A5>B5	"	9,11,14,29, 30	13	"	"	"	"	"	"	"

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* TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
P78	A4>B4	Propagation delay time (t_{PLH}), pulse on B, 5 V on A, C_L = 130 pF	9,11,12,13, 14,29,30	31	Open	28	Open	5	55	ns	13,14
P79	A3>B3	"	9,11,12,13, 14,23,29, 30,31	21	"	"	"	"	"	"	"
P80	A2>B2	"	9,11,12,13, 14,21,23, 24,29,30,31	4	"	"	"	"	"	"	"
P81	A1>B1	"	4,5,9,11, 12,13,14, 21,23,24, 29,30,31	25	"	"	"	"	"	"	"
P82	A0>B0	"	4,5,6,9,11, 12,13,14, 21,23,24, 25,29,30,31	26	"	"	"	"	"	"	"
P83	A7=B7	"	29	9	"	8	"	"	65	"	"
P84	A6=B6	"	9,11,29	30	"	"	"	"	"	"	"
P85	A5=B5	"	9,11,14,29, 30	13	"	"	"	"	"	"	"
P86	A4=B4	"	9,11,12,13, 14,29,30	31	"	"	"	"	"	"	"
P87	A3=B3	"	9,11,12,13, 14,23,29, 30,31	21	"	"	"	"	"	"	"
P88	A2=B2	"	9,11,12,13, 14,21,23, 24,29,30,31	4	"	"	"	"	"	"	"

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* TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
P89	A1=B1	Propagation delay time (t_{PLH}), pulse on B, 5 V on A, C_L = 130 pF	4,5,9,11, 12,13,14, 21,23,24, 29,30,31	25	Open	8	Open	5	65	ns	13,14
P90	A0=B0	Propagation delay time (t_{PLH}), pulse on B, 5 V on A, C_L = 130 pF	4,5,6,9,11, 12,13,14, 21,23,24, 25,29,30,31	26	"	8	Open	"	65	"	"
P91	A7<B7	Propagation delay time (t_{PLH}), pulse on B, 0 V on A, C_L = 130 pF	Open	9	"	7	29	"	55	"	"
P92	A6<B6	"	"	30	"	"	9,11,29	"	"	"	"
P93	A5<B5	"	"	13	"	"	9,11,14,29, 30	"	"	"	"
P94	A4<B4	"	"	31	"	"	9,11,12,13, 14,29,30	"	"	"	"
P95	A3<B3	"	"	21	"	"	9,11,12,13, 14,23,29, 30,31	"	"	"	"
P96	A2<B2	"	"	4	"	"	9,11,12,13, 14,21,23, 24,29,30,31	"	"	"	"
P97	A1<B1	"	"	25	"	"	4,5,9,11, 12,13,14, 21,23,24, 29,30,31	"	"	"	"
P98	A0<B0	"	"	26	"	"	4,5,6,9,11, 12,13,14, 21,23,24, 25,29,30,31	"	"	"	"
P99	A7=B7	"	"	9	"	8	29	"	65	"	"

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* TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
P100	A6=B6	Propagation delay time (t_{PLH}), pulse on B, 0 V on A, C_L = 130 pF	Open	30	Open	8	9,11,29	5	65	ns	13,14
P101	A5=B5	"	"	13	"	"	9,11,14,29, 30	"	"	"	"
P102	A4=B4	"	"	31	"	"	9,11,12,13, 14,29,30	"	"	"	"
P103	A3=B3	"	"	21	"	"	9,11,12,13, 14,23,29, 30,31	"	"	"	"
P104	A2=B2	"	"	4	"	"	9,11,12,13, 14,21,23, 24,29,30,31	"	"	"	"
P105	A1=B1	"	"	25	"	"	4,5,9,11, 12,13,14, 21,23,24, 29,30,31	"	"	"	"
P106	A0=B0	"	"	26	"	"	4,5,6,9,11, 12,13,14, 21,23,24, 25,29,30,31	"	"	"	"
Q1	1A<B IN to 1A>B OUT	Propagation delay time (t_{PHL}), C_L = 130 pF	"	2	3	28	4,5,6,9,11, 12,13,14,21, 22,23,24,25, 26,29,30,31	"	45	"	"
Q2	1A=B IN to 1A>B OUT	"	"	22	3	28	2	"	"	"	"
Q3	1A>B IN to 1A<B OUT	"	"	3	2	7	22	"	"	"	"
Q4	1A=B IN to 1A<B OUT	"	"	22	2	7	3	"	"	"	"

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* TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q5	1A=B IN to 1A=B OUT	Propagation delay time (t_{PHL}), C_L = 130 pF	Open	22	2	8	3,4,5,6,9, 11,12,13,14, 21,23,24,25, 26,29,30,31	5	50	ns	13,14
Q6	2A<B IN to 2A>B OUT	"	"	36	15	34	16,17,18,19, 20,35,37,38, 39	"	25	"	"
Q7	2A=B IN to 2A>B OUT	"	"	35	15	34	36	"	"	"	"
Q8	2A>B IN to 2A<B OUT	"	"	15	36	32	35	"	"	"	"
Q9	2A=B IN to 2A<B OUT	"	"	35	"	32	15	"	"	"	"
Q10	2A=B IN to 2A=B OUT	"	"	35	"	33	15,16,17,18, 19,20,37,38, 39	"	30	"	"
Q11	A11>B11	Propagation delay time (t_{PHL}), pulse on A, 0 V on B, C_L = 130 pF	"	20	Open	34	16	"	40	"	"
Q12	A10>B10	"	"	39	"	"	16,19,20	"	"	"	"
Q13	A9>B9	"	"	18	"	"	16,19,20,38, 39	"	"	"	"
Q14	A8>B8	"	"	17	"	"	16,18,19,20, 37,38,39	"	"	"	"
Q15	A11=B11	"	"	20	"	33	16	"	"	"	"
Q16	A10=B10	"	"	39	"	"	16,19,20	"	"	"	"
Q17	A9=B9	"	"	18	"	"	16,19,20, 38,39	"	"	"	"

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* TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q18	A8=B8 Propagation delay time (t_{PHL}), pulse on A, 0 V on B, C_L = 130 pF	Open	17	Open	33	16,18,19,20, 37,38,39	5	40	ns	13,14
Q19	A11<B11 Propagation delay time (t_{PHL}), pulse on A, 5 V on B, C_L = 130 pF	16	20	"	32	Open	"	"	"	"
Q20	A10<B10 "	16,19,20	39	"	"	"	"	"	"	"
Q21	A9<B9 "	16,19,20,38, 39	18	"	"	"	"	"	"	"
Q22	A8<B8 "	16,18,19,20, 37,38,39	17	"	"	"	"	"	"	"
Q23	A11=B11 "	16	20	"	33	"	"	"	"	"
Q24	A10=B10 "	16,19,20	39	"	"	"	"	"	"	"
Q25	A9=B9 "	16,19,20,38, 39	18	"	"	"	"	"	"	"
Q26	A8=B8 "	16,18,19,20, 37,38,39	17	"	"	"	"	"	"	"
Q27	A11>B11 Propagation delay time (t_{PHL}), pulse on B, 5 V on A, C_L = 130 pF	20	16	"	34	"	"	"	"	"
Q28	A10>B10 "	16,20,39	19	"	"	"	"	"	"	"
Q29	A9>B9 "	16,18,19,20, 39	38	"	"	"	"	"	"	"
Q30	A8>B8 "	16,17,18,19, 20,38,39	37	"	"	"	"	"	"	"
Q31	A11=B11 "	20	16	"	33	"	"	"	"	"
Q32	A10=B10 "	16,20,39	19	"	33	"	"	"	"	"

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* TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q33	A9=B9	Propagation delay time (t_{PHL}), pulse on B, 5 V on A, C_L = 130 pF	16,18,19,20, 39	38	Open	33	Open	5	40	ns	13,14
Q34	A8=B8	Propagation delay time (t_{PHL}), pulse on B, 5 V on A, C_L = 130 pF	16,17,18,19, 20,38,39	37	"	33	Open	"	"	"	"
Q35	A11<B11	Propagation delay time (t_{PHL}), pulse on B, 0 V on A, C_L = 130 pF	Open	16	"	32	20	"	"	"	"
Q36	A10<B10	"	"	19	"	"	16,20,39	"	"	"	"
Q37	A9<B9	"	"	38	"	"	16,18,19,20, 39	"	"	"	"
Q38	A8<B8	"	"	37	"	"	16,17,18,19, 20,38,39	"	"	"	"
Q39	A11=B11	"	"	16	"	33	20	"	"	"	"
Q40	A10=B10	"	"	19	"	"	16,20,39	"	"	"	"
Q41	A9=B9	"	"	38	"	"	16,18,19,20, 39	"	"	"	"
Q42	A8=B8	"	"	37	"	"	16,17,18,19, 20,38,39	"	"	"	"
Q43	A7>B7	Propagation delay time (t_{PHL}), pulse on A, 0 V on B, C_L = 130 pF	"	29	"	28	9	"	60	"	"
Q44	A6>B6	"	"	11	"	"	9,29,30	"	"	"	"
Q45	A5>B5	"	"	14	"	"	9,11,13,29, 30	"	"	"	"
Q46	A4>B4	"	"	12	"	"	9,11,13,14, 29,30,31	"	"	"	"
Q47	A3>B3	"	"	23	"	"	9,11,12,13, 14,21,29, 30,31	"	"	"	"

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* TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q48	A2>B2	Propagation delay time (t_{PHL}), pulse on A, 0 V on B, C_L = 130 pF	Open	24	Open	28	4,9,11,12, 13,14,21,23, 29,30,31	5	60	ns	13,14
Q49	A1>B1	"	"	5	"	"	4,9,11,12, 13,14,21,23, 24,25,29,30, 31	"	"	"	"
Q50	A0>B0	"	"	6	"	"	4,5,9,11,12, 13,14,21,23, 24,25,26,29, 30,31	"	"	"	"
Q51	A7=B7	"	"	29	"	8	9	"	"	"	"
Q52	A6=B6	"	"	11	"	"	9,29,30	"	"	"	"
Q53	A5=B5	"	"	14	"	"	9,11,13,29, 30	"	"	"	"
Q54	A4=B4	"	"	12	"	"	9,11,13,14, 29,30,31	"	"	"	"
Q55	A3=B3	"	"	23	"	"	9,11,12,13, 14,21,29,30, 31	"	"	"	"
Q56	A2=B2	"	"	24	"	"	4,9,11,12, 13,14,21,23, 29,30,31	"	"	"	"
Q57	A1=B1	"	"	5	"	"	4,9,11,12, 13,14,21,23, 24,25,29,30, 31	"	"	"	"
Q58	A0=B0	"	"	6	"	"	4,5,9,11,12, 13,14,21,23, 24,25,26,29, 30,31	"	"	"	"

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* TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q59	A7<B7	Propagation delay time (t_{PHL}), pulse on A, 5 V on B, C_L = 130 pF	9	29	Open	7	Open	5	60	ns	13,14
Q60	A6<B6	"	9,29,30	11	"	"	"	"	"	"	"
Q61	A5<B5	"	9,11,13,29, 30	14	"	"	"	"	"	"	"
Q62	A4<B4	"	9,11,13,14, 29,30,31	12	"	"	"	"	"	"	"
Q63	A3<B3	"	9,11,12,13, 14,21,29,30, 31	23	"	"	"	"	"	"	"
Q64	A2<B2	"	4,9,11,12, 13,14,21,23, 29,30,31	24	"	"	"	"	"	"	"
Q65	A1<B1	"	4,9,11,12, 13,14,21,23, 24,25,29,30, 31	5	"	"	"	"	"	"	"
Q66	A0<B0	"	4,5,9,11,12, 13,14,21,23, 24,25,26,29, 30,31	6	"	"	"	"	"	"	"
Q67	A7=B7	"	9	29	"	8	"	"	"	"	"
Q68	A6=B6	"	9,29,30	11	"	"	"	"	"	"	"
Q69	A5=B5	"	9,11,13,29, 30	14	"	"	"	"	"	"	"
Q70	A4=B4	"	9,11,13,14, 29,30,31	12	"	"	"	"	"	"	"

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* TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q71	A3=B3	Propagation delay time (t_{PHL}), pulse on A, 5 V on B, C_L = 130 pF	9,11,12,13, 14,21,29, 30,31	23	Open	8	Open	5	60	ns	13,14
Q72	A2=B2	"	4,9,11,12, 13,14,21,23, 29,30,31	24	"	"	"	"	"	"	"
Q73	A1=B1	"	4,9,11,12, 13,14,21, 23,24,25, 29,30,31	5	"	"	"	"	"	"	"
Q74	A0=B0	"	4,5,9,11, 12,13,14, 21,23,24, 25,26,29, 30,31	6	"	"	"	"	"	"	"
Q75	A7>B7	Propagation delay time (t_{PHL}), pulse on B, 5 V on A, C_L = 130 pF	29	9	"	28	"	"	"	"	"
Q76	A6>B6	"	9,11,29	30	"	"	"	"	"	"	"
Q77	A5>B5	"	9,11,14,29, 30	13	"	"	"	"	"	"	"
Q78	A4>B4	"	9,11,12,13, 14,29,30	31	"	"	"	"	"	"	"
Q79	A3>B3	"	9,11,12,13, 14,23,29, 30,31	21	"	"	"	"	"	"	"
Q80	A2>B2	"	9,11,12,13, 14,21,23, 24,29,30,31	4	"	"	"	"	"	"	"
Q81	A1>B1	"	4,5,9,11, 12,13,14, 21,23,24, 29,30,31	25	"	"	"	"	"	"	"

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* TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q82	A0>B0	Propagation delay time (t_{PHL}), pulse on B, 5 V on A, C_L = 130 pF	4,5,6,9,11, 12,13,14, 21,23,24, 25,29,30,31	26	Open	28	Open	5	60	ns	13,14
Q83	A7=B7	"	29	9	"	8	"	"	"	"	"
Q84	A6=B6	"	9,11,29	30	"	"	"	"	"	"	"
Q85	A5=B5	"	9,11,14,29, 30	13	"	"	"	"	"	"	"
Q86	A4=B4	"	9,11,12,13, 14,29,30	31	"	"	"	"	"	"	"
Q87	A3=B3	"	9,11,12,13, 14,23,29, 30,31	21	"	"	"	"	"	"	"
Q88	A2=B2	"	9,11,12,13, 14,21,23, 24,29,30,31	4	"	"	"	"	"	"	"
Q89	A1=B1	"	4,5,9,11, 12,13,14, 21,23,24, 29,30,31	25	"	"	"	"	"	"	"
Q90	A0=B0	"	4,5,6,9,11, 12,13,14, 21,23,24, 25,29,30,31	26	"	"	"	"	"	"	"
Q91	A7<B7	Propagation delay time (t_{PHL}), pulse on B, 0 V on A, C_L = 130 pF	Open	9	"	7	29	"	"	"	"
Q92	A6<B6	"	"	30	"	"	9,11,29	"	"	"	"
Q93	A5<B5	"	"	13	"	"	9,11,14,29, 30	"	"	"	"

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* TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

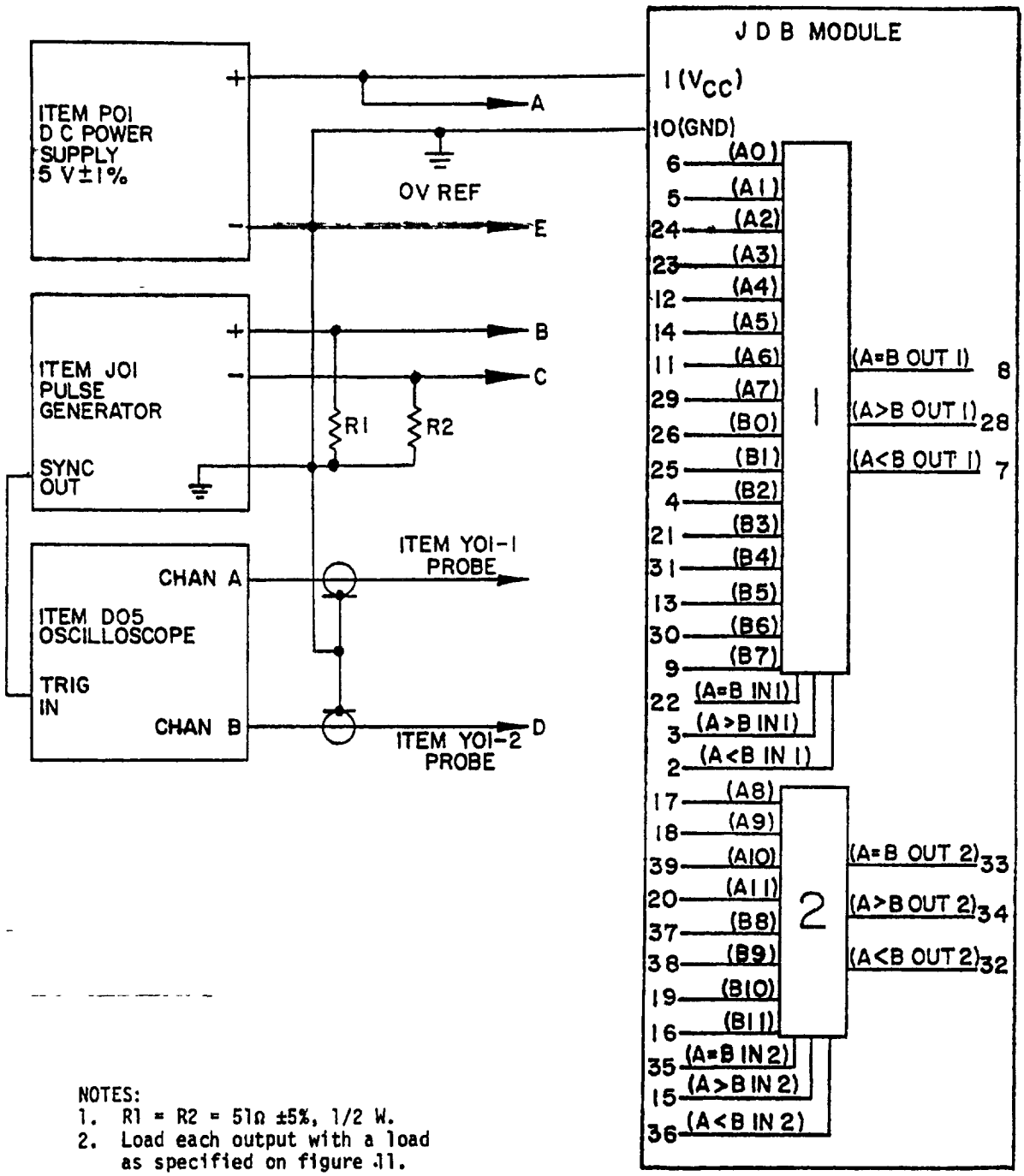
Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q94	A4<B4 Propagation delay time (t_{PHL}), pulse on B, 0 V on A, C_L = 130 pF	Open	31	Open	7	9,11,12,13, 14,29,30	5	60	ns	13,14
Q95	A3<B3 "	"	21	"	"	9,11,12,13, 14,23,29, 30,31	"	"	"	"
Q96	A2<B2 "	"	4	"	"	9,11,12,13, 14,21,23, 24,29,30,31	"	"	"	"
Q97	A1<B1 "	"	25	"	"	4,5,9,11, 12,13,14, 21,23,24, 29,30,31	"	"	"	"
Q98	A0<B0 "	"	26	"	"	4,5,6,9,11, 12,13,14, 21,23,24, 25,29,30,31	"	"	"	"
Q99	A7=B7 "	"	9	"	8	29	"	"	"	"
Q100	A6=B6 "	"	30	"	"	9,11,29	"	"	"	"
Q101	A5=B5 "	"	13	"	"	9,11,14,29, 30	"	"	"	"
Q102	A4=B4 "	"	31	"	"	9,11,12,13, 14,29,30	"	"	"	"
Q103	A3=B3 "	"	21	"	"	9,11,12,13, 14,23,29, 30,31	"	"	"	"
Q104	A2=B2 "	"	4	"	"	9,11,12,13, 14,21,23, 24,29,30,31	"	"	"	"

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* TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q105	A1=B1	Propagation delay time (t_{PHL}), pulse on B, 0 V on A, C_L = 130 pF	Open	25	Open	8	4,5,9,11, 12,13,14, 21,23,24, 29,30,31	5	60	ns	13,14
Q106	A0=B0	Propagation delay time (t_{PHL}), pulse on B, 0 V on A, C_L = 130 pF	Open	26	Open	8	4,5,6,9,11, 12,13,14, 21,23,24, 25,29,30,31	5	60	ns	13,14

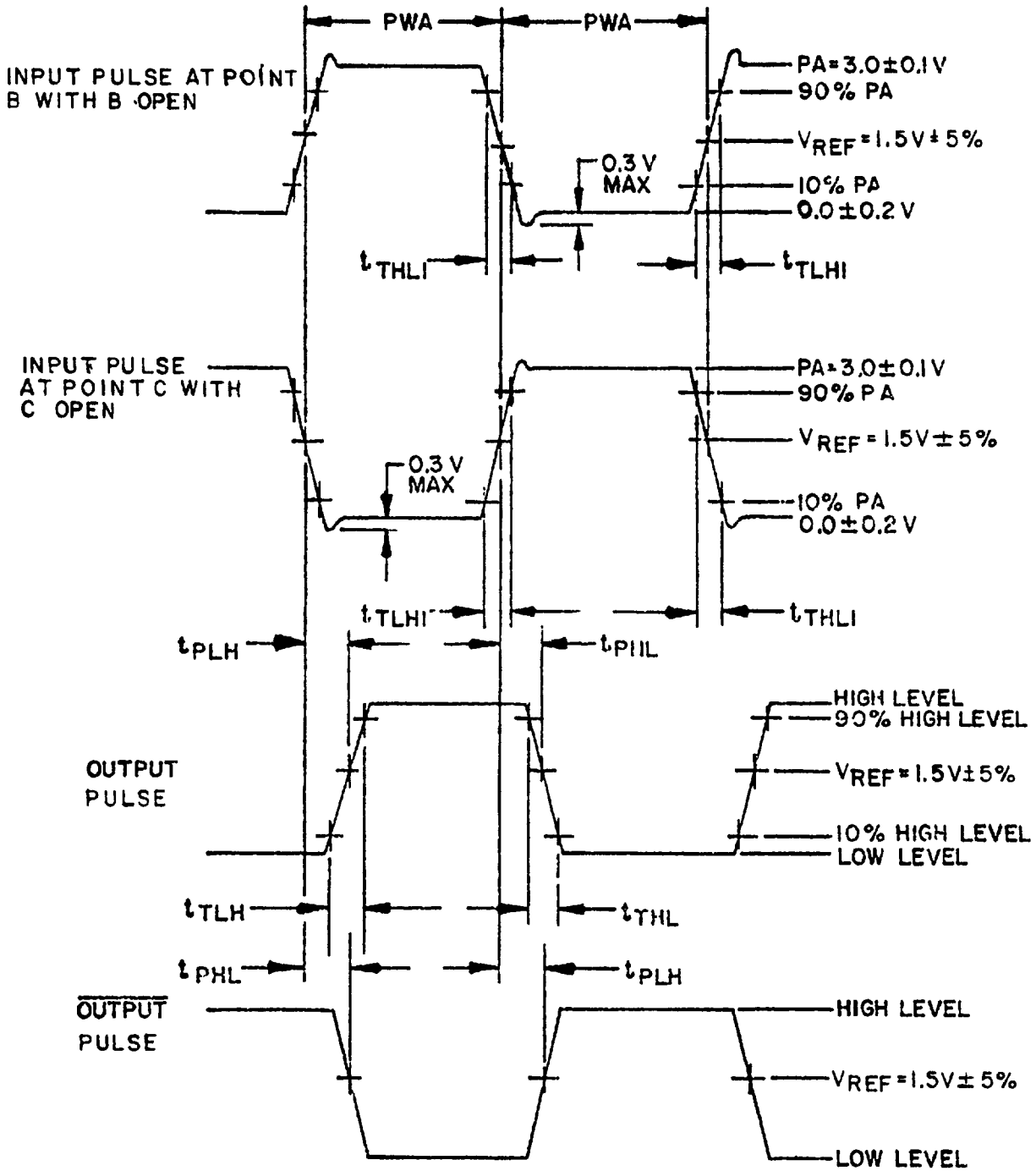
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load each output with a load as specified on figure 11.

* FIGURE 13. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables X and XI.

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NOTES:
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   $t_{TLHI} = t_{THLI}$ = 10 ±2 ns.
2. Before making measurements, the output levels shall be verified to be $\leq$ 0.4 V for the low level and $\geq$ 2.4 V for the high level.

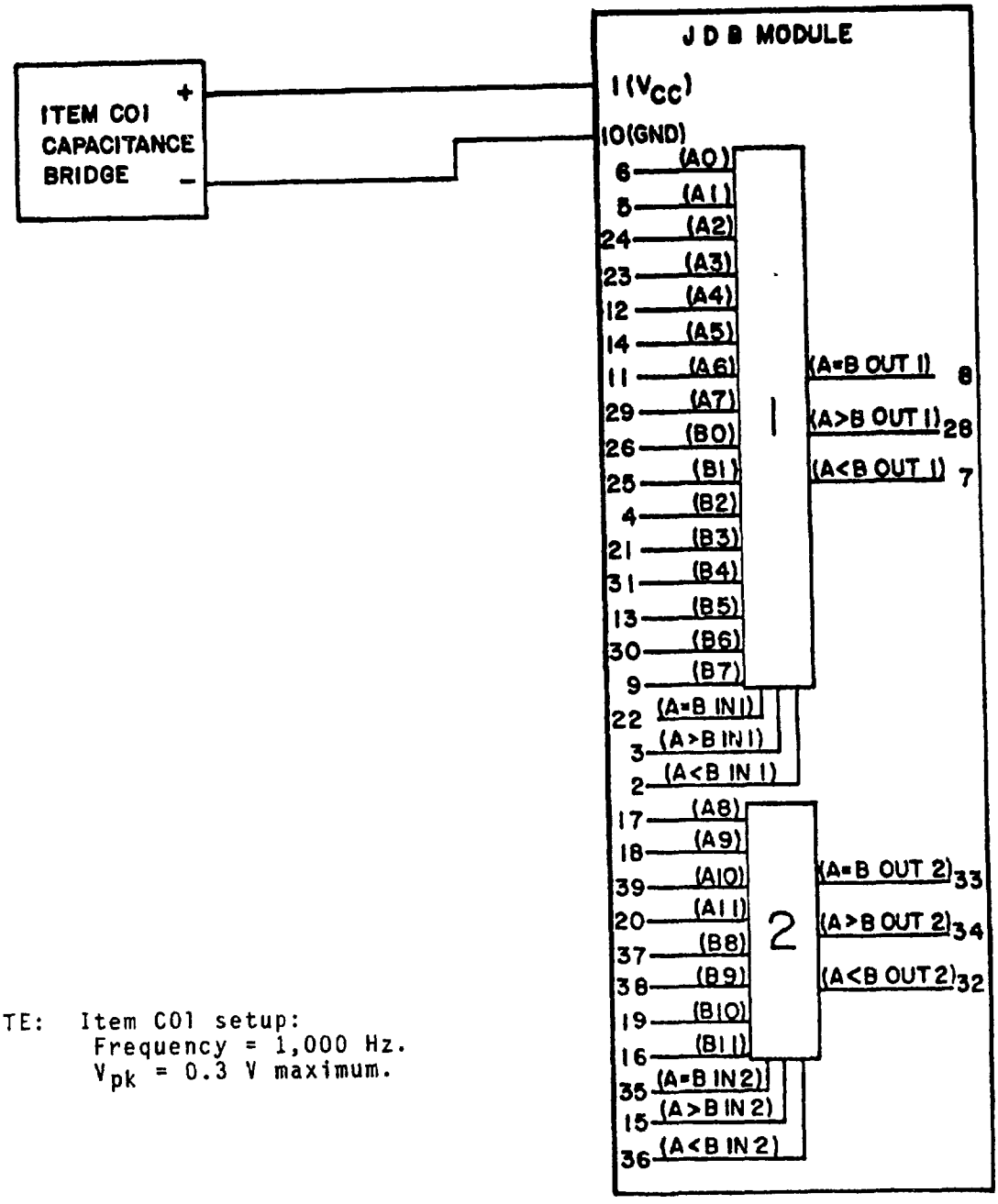
* FIGURE 14. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables X and XI.

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TABLE XII. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
R	Power supply filter capacitance	0.005	0.015	µF	15

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NOTE: Item C01 setup:
Frequency = 1,000 Hz.
V_{pk} = 0.3 V maximum.

FIGURE 15. Power supply filter capacitance test circuit for table XII.

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* TABLE XIII. Output crosstalk (high-level and low-level) test.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
S1	1A>B OUT	Output crosstalk, high level	3	2,15, 22	35	36	28	0	400	mV	16,17
S2	1A<B OUT	"	2	3,22, 36	35	15	7	"	"	"	"
S3	1A=B OUT	"	22	2,3,35	36	15	8	"	"	"	"
S4	2A>B OUT	"	15	3,35, 36	22	2	34	"	"	"	"
S5	2A<B OUT	"	36	2,15, 35	22	3	32	"	"	"	"
S6	2A=B OUT	"	35	15,22, 36	2	3	33	"	"	"	"
T1	1A>B OUT	Output crosstalk, low level	22	2,3,15	35	36	28	"	"	"	"
T2	1A<B OUT	"	22	2,3,36	35	15	7	"	"	"	"
T3	1A=B OUT	"	2	3,22, 35	36	15	8	"	"	"	"
T4	2A>B OUT	"	35	15,3, 36	22	2	34	"	"	"	"
T5	2A<B OUT	"	35	2,15, 36	22	3	32	"	"	"	"
T6	2A=B OUT	"	36	15,22, 35	2	3	33	"	"	"	"

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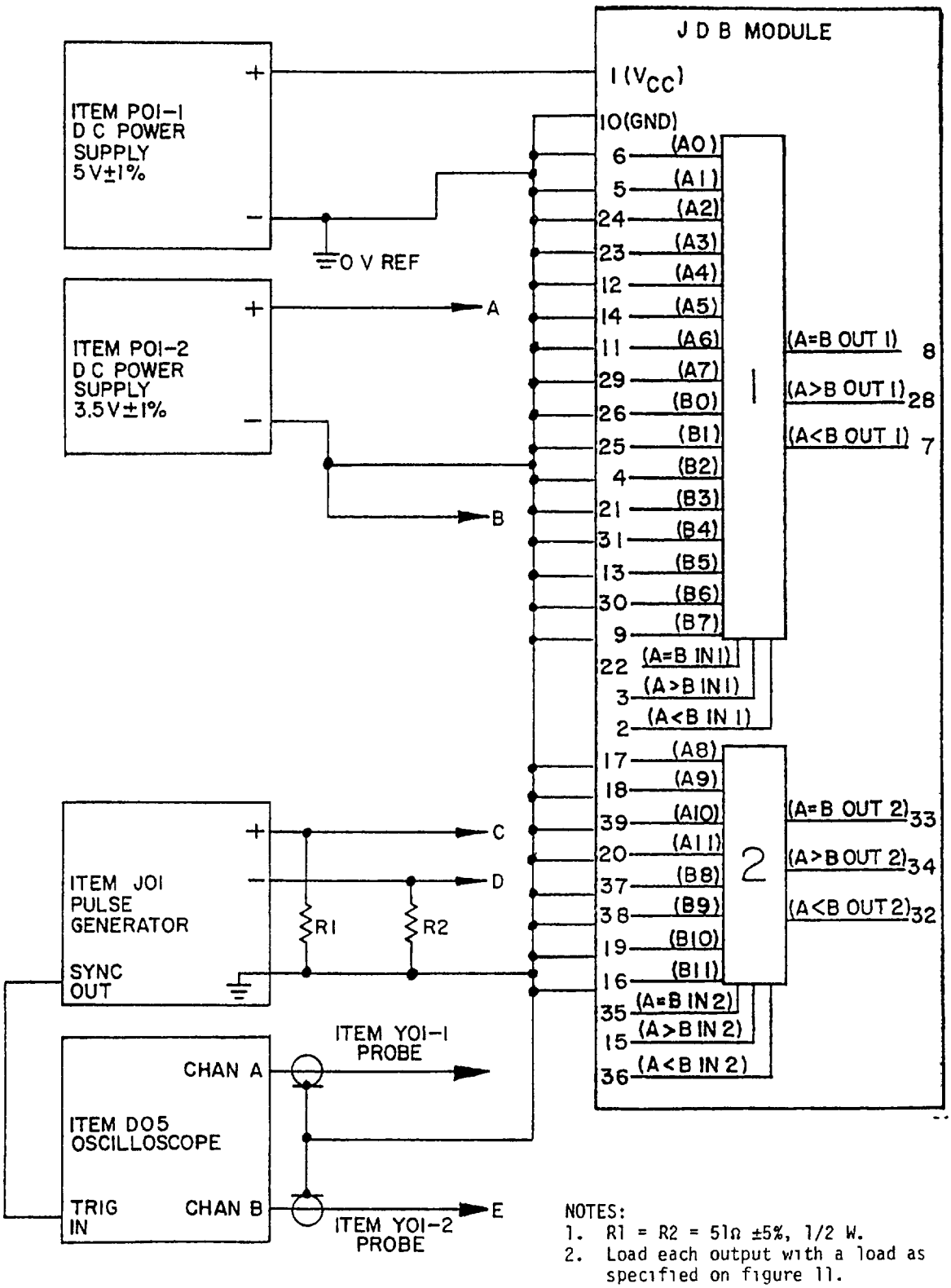
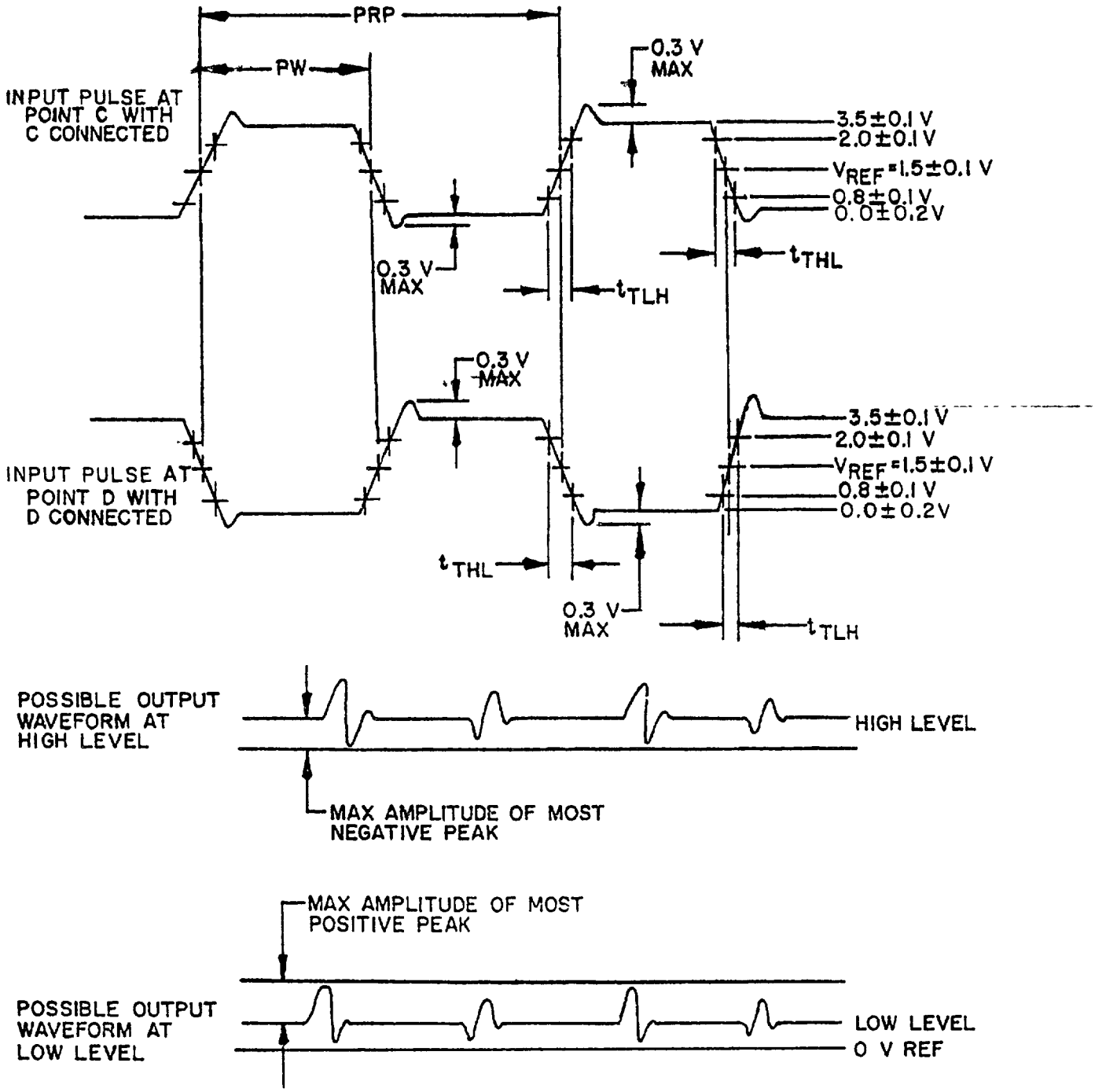


FIGURE 16. Output crosstalk (high-level and low-level) test circuit for table XIII.

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NOTE:
1. Input pulse parameters:
   (a) $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at all input pins).
   (b) PW = 500 ±50 ns.
   (c) PRP = 1,000 ±100 ns.
2. The lead length of electrical conductors between resistor R1 and input pins, probes Y01-1 and Y01-2 to input and output pins, load circuits to output pins, and ground leads of probes Y01-1 and Y01-2 to test fixture and module ground pins shall be less than 2 inches (50.8 mm).

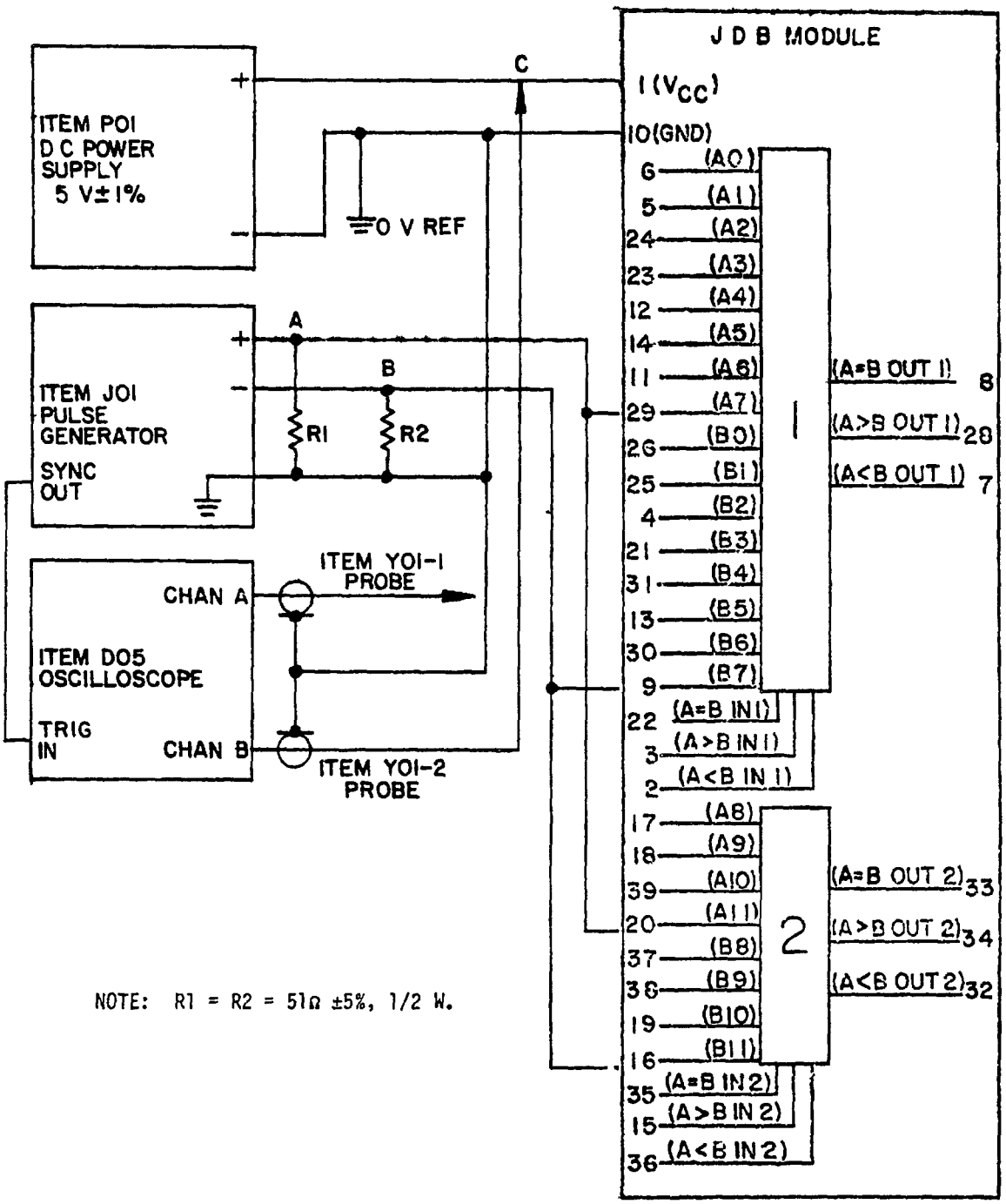
FIGURE 17. Output crosstalk (high-level and low-level) test circuit waveforms for table XIII.

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TABLE XIV. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
U	Power supply transient amplitude	0	400	mV	18,19

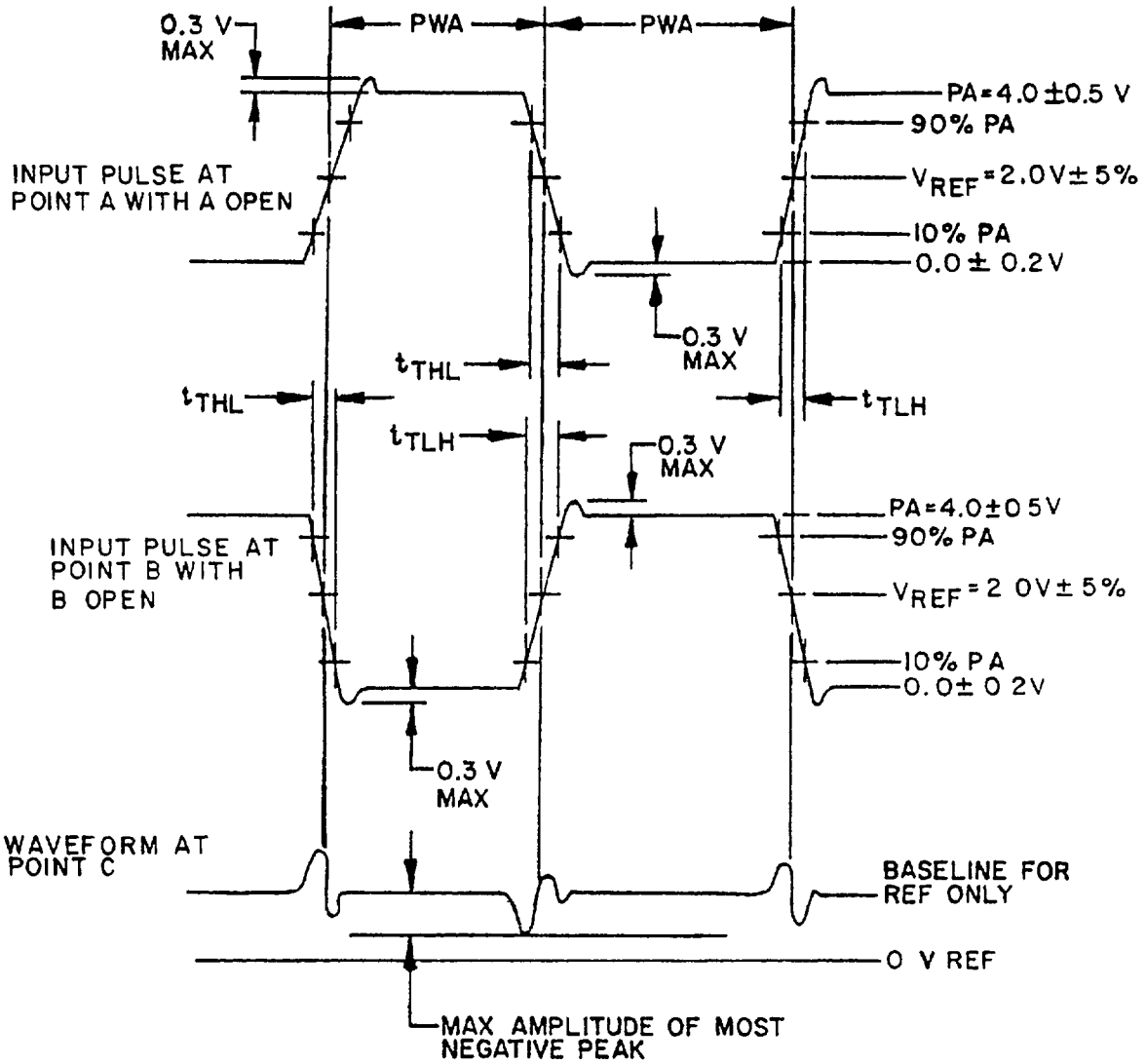
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NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

FIGURE 18. Power supply transient amplitude test circuit for table XIV.

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NOTE: Input pulse parameters:
PWA = 500 ±50 ns.
$t_{TLH} = t_{THL} = 10 \pm 2$ ns.

* FIGURE 19. Power supply transient amplitude test circuit waveforms for table XIV.

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TABLE XV. Input capacitance test.

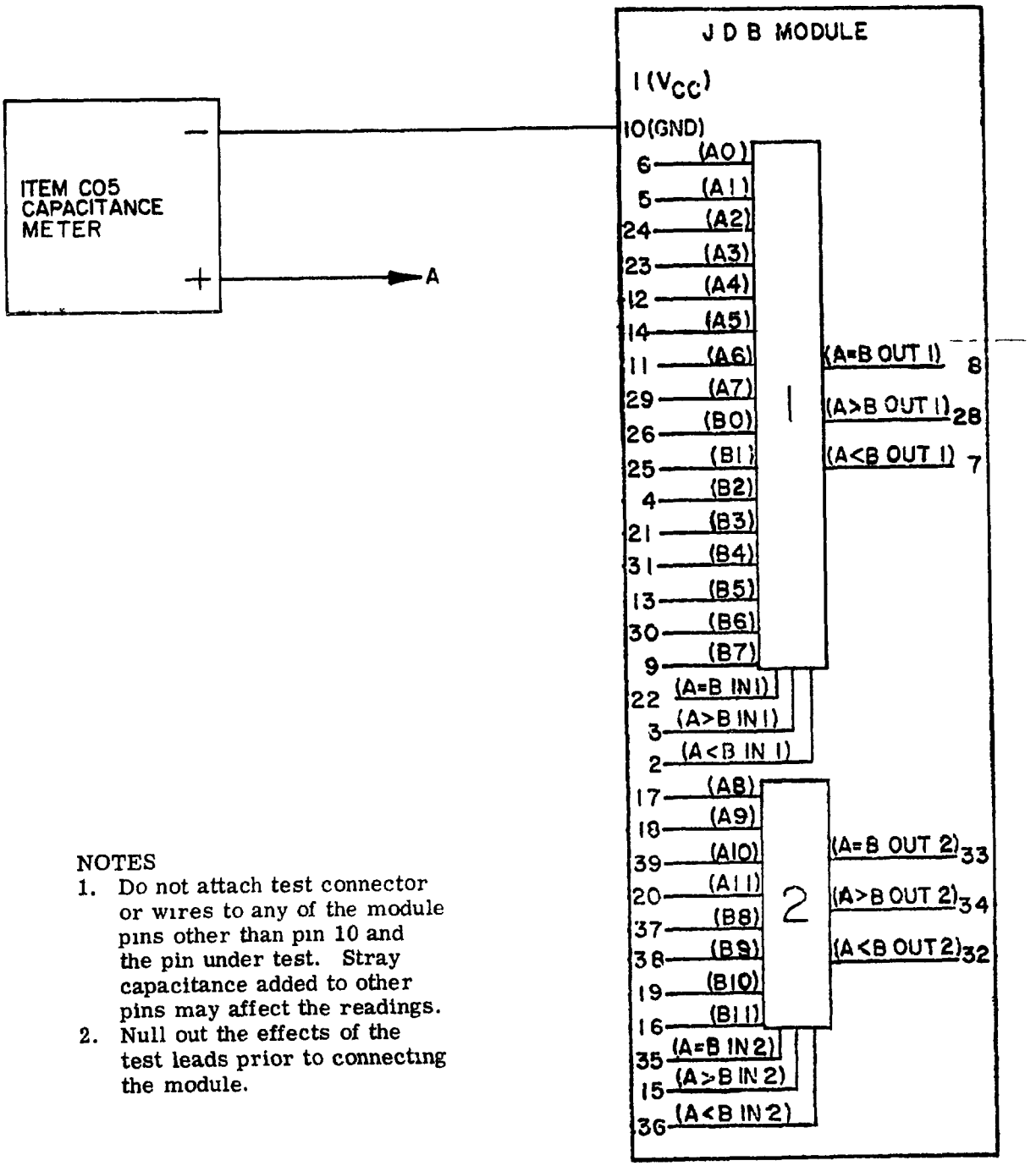
Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
V1	1A>B IN	Input capacitance	3	2	15	pF	20
V2	1A<B IN	"	2	"	"	"	"
V3	2A>B IN	"	15	"	"	"	"
V4	2A<B IN	"	36	"	"	"	"
V5	1A=B IN	"	22	"	"	"	"
V6	2A=B IN	"	35	"	"	"	"
V7	A0	"	6	"	"	"	"
V8	A1	"	5	"	"	"	"
V9	A2	"	24	"	"	"	"
V10	A3	"	23	"	"	"	"
V11	A4	"	12	"	"	"	"
V12	A5	"	14	"	"	"	"
V13	A6	"	11	"	"	"	"
V14	A7	"	29	"	"	"	"
V15	A8	"	17	"	"	"	"
V16	A9	"	18	"	"	"	"
V17	A10	"	39	"	"	"	"
V18	A11	"	20	"	"	"	"
V19	B0	"	26	"	"	"	"
V20	B1	"	25	"	"	"	"

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TABLE XV. Input capacitance test - Continued.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
V21	B2	Input capacitance	4	2	15	pF	20
V22	B3	"	21	"	"	"	"
V23	B4	"	31	"	"	"	"
V24	B5	"	13	"	"	"	"
V25	B6	"	30	"	"	"	"
V26	B7	"	9	"	"	"	"
V27	B8	"	37	"	"	"	"
V28	B9	"	38	"	"	"	"
V29	B10	"	19	"	"	"	"
V30	B11	"	16	"	"	"	"

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NOTES
1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.

* FIGURE 20. Input capacitance test circuit for table XV.

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